

॥ नवोन्मेषेण देशनिर्माणम् ॥

భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
भारतीय प्रौद्योगिकी संस्थान हैदराबाद
Indian Institute of Technology Hyderabad

MASooda: A Hierarchical, Priority-Aware and Resilient Multi-Agent System for Urban Air Traffic Management

Group Design Project Report

Team Name: MASooda

Team Members:

1. R Lalith Vishnu
2. Sampadram
3. Mohammed Abdur Rehman
4. Bhuvan
5. Tanmay Banjari
6. Dhadheechi

AI3403 - Multi-Agent Systems
Department of Artificial Intelligence
Indian Institute of Technology Hyderabad

Kandi, Telangana, INDIA - November 2026

MASooda: A Hierarchical, Priority-Aware and Resilient Multi-Agent System for Urban Air Traffic Management

Group Design Project Report

Team Name: MASooda

Team Members:

1. **R Lalith Vishnu**
2. **Sampadram**
3. **Mohammed Abdur Rehman**
4. **Bhuvan**
5. **Tanmay Banjari**
6. **Dhadheechi**

Project Supervisor: Dr. Venkatraman Renganathan
Assistant Professor, IITH

AI3403 - Multi-Agent Systems
Department of Artificial Intelligence
Indian Institute of Technology Hyderabad

GDP: IITH-AI-MAS-2026-#0001

Kandi, Telangana, INDIA - November 2026



॥ नवोन्मेषेण देशनिर्माणम् ॥

భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్

भारतीय प्रौद्योगिकी संस्थान हैदराबाद

Indian Institute of Technology Hyderabad

Team Name: MASooda

GDP: IITH-AI-MAS-2026-#0001

Copyright © 2026

IIT Hyderabad.

Abstract

Dense urban drone traffic needs a traffic manager that keeps neighbouring aircraft apart within fractions of a second, plans the airspace as a whole, and keeps working when its links fail or some aircraft broadcast false positions. This report presents MA-Sooda, a two-tier multi-agent system for urban air traffic management. In the local tier, drones follow their corridors with Boids-style rules and resolve short-horizon conflicts with priority-weighted consensus ADMM. In the global tier, a Drone Traffic Controller (DTC) allocates corridors and altitude layers by priority and capacity, reserves an emergency layer and falls back to cached directives when its link fails. Position spoofing is countered by a trusted-range check and W-MSR filtering, and an online Signal Temporal Logic monitor measures the minimum-separation margin every tick. The core system was extended into a live, map-constrained airspace on IIT Hyderabad campus and Hyderabad roads with seven layers, V2V layer negotiation with DTC congestion pricing, a battery, BMS and thermal model for every drone, an operator console and ten scripted scenarios.

In a 141-run campaign, ADMM converged in about ten iterations regardless of fleet size, at under 2 ms per tick for 75 agents. The campaign also revealed a horizontal-only safety metric that inflates the hierarchical violation counts, which the report documents. In the layer-aware live airspace, 29 of 30 verification runs with up to 90 drones had no loss of separation, and layer agreement raised arrivals by 36% and cut ADMM conflicts by 71% with 300 drones, without near misses. The scripted scenarios demonstrate emergency right of way, lost-link policies, spoofer quarantine and health-driven diversions without energy depletion.

Keywords: Multi-agent systems, urban air mobility, UTM, hierarchical control, ADMM, Boids, resilient consensus, W-MSR, signal temporal logic, battery management system.

Contents

List of Figures iv

List of Tables vi

Acronyms vi

1 Introduction **2**

1.1 Vision & Motivation . . . 2
1.2 Goals . . . 3
1.3 Scope . . . 4
1.4 Contributions of this Work . . . 4
1.5 Organisation of the Report . . . 5

2 Literature Review **6**

2.1 Traffic Management Concepts for Low-Altitude Airspace . . . 6
2.2 Local Coordination: Flocking, Platooning and Car-Following . . . 7
2.3 Distributed Optimisation for Conflict Resolution . . . 7
2.4 Hierarchical versus Centralised and Decentralised Control . . . 8
2.5 Resilient Consensus under Adversarial Information . . . 8
2.6 Formal Runtime Verification with Signal Temporal Logic . . . 8
2.7 Battery Health Management for Electric Aircraft . . . 8
2.8 Research Gap . . . 9

3 Preliminaries **11**

3.1 Notations . . . 11
3.1.1 Agents, Time and Airspace . . . 11
3.1.2 Norms, Sets and Operators . . . 11
3.2 Background on Preliminaries . . . 11
3.2.1 2.5D Kinematics . . . 11
3.2.2 Communication Graphs and Network Measures . . . 12
3.2.3 Boids Rules . . . 13
3.2.4 The Alternating Direction Method of Multipliers . . . 13
3.2.5 Resilient Consensus and W-MSR . . . 13
3.2.6 Signal Temporal Logic and Robustness . . . 14
3.2.7 Rotor Power and Battery Equivalent Circuit . . . 14

4 Problem Formulation **15**
4.1 Problem Statement . . . 15
4.1.1 Execution Phase #1: Corridor Benchmark . . . 16
4.1.2 Execution Phase #2: Map-Constrained Live Airspace . . . 16
4.2 Solution Approach . . . 17
4.2.1 Architecture Overview . . . 17
4.2.2 Per-Tick Pipeline . . . 18
4.2.3 Local Tier: Corridor Following, Boids and Car-Following . . . 19
4.2.4 Local Tier: Priority-Weighted Consensus ADMM . . . 20
4.2.5 Global Tier: Drone Traffic Controller . . . 21
4.2.6 Layer Agreement: V2V Negotiation with DTC Pricing . . . 22
4.2.7 Resilience: Attack Model and Defence . . . 23
4.2.8 Verification: Online STL Monitoring . . . 24
4.2.9 Drone Health Model . . . 24
4.2.10 Operator Console and Scenario Engine . . . 26
4.2.11 Architectures Compared . . . 26
4.2.12 Theoretical Analysis . . . 27
4.2.13 Complexity Analysis . . . 28

5 Results **29**
5.1 Simulation Results . . . 29
5.1.1 Simulation Setup . . . 29
5.1.2 Results & Discussions . . . 31
5.2 Limitations . . . 42

6 Conclusion **44**
6.1 Project Summary . . . 44
6.2 Future Outlook . . . 45

7 Contributions **47**
7.1 Individual Contributions . . . 47

References 50

Appendices

A Configuration Parameters **55**

B Scripted Scenarios **58**

List of Figures

4.1 Corridor benchmark world (60 agents, seed 42, $t = 60\,\mathrm{s}$): three corridors with entry, merge and exit zones, V2V links of radius 40 m, and the DTC sending corridor and altitude directives every 2 s. 16

4.2 IIT Hyderabad campus map used to trace the corridor network, the geofence and the 13 vertiports of the live airspace. 17

4.3 The MASooda architecture. The DTC sets slow, airspace-wide references; the drones resolve fast, local interactions among themselves. Filtered reports from the resilience layer and health states feed the local tier, and the verification layer observes the outcome independently. 18

4.4 Order of operations within one 0.1 s tick of the map-constrained engine. The corridor benchmark uses the same order without the map-specific steps. . . . 19

4.5 Algorithm 1: priority-aware corridor and layer allocation performed by the DTC at every update. 22

5.1 Side view of the corridor benchmark (hierarchical, 60 agents, $t = 60\,\mathrm{s}$). Emergencies occupy the reserved layer 2, and delivery agents are spread over layers 0 and 1. Pairs stacked on different layers have near-zero horizontal distance. . 33

5.2 E1: ADMM effort per solve is flat in N; total time per tick grows roughly linearly with the number of conflicts. 34

5.3 E1: converged fraction and mean iterations versus the ADMM iteration limit at $N = 60$. Convergence saturates at about 20 iterations. 34

5.4 E2: throughput and corridor utilisation in the corridor benchmark. 35

5.5 E3: progress and path of an emergency agent and of the same agent as an ordinary delivery drone in the paired control run ($N = 60$, seed 42). 36

5.6 E4: throughput, violating ticks, mean recovery time and minimum robustness for the three architectures, attack off and on (25 m offset). Points are individual seeds. 37

5.7 E4: STL robustness over time, mean and 95% band across five seeds. The hierarchical curve drops below zero within about 2.5 s, before any merge or attack, which reveals the cross-layer artefact. 38

5.8 E5: fraction of ticks with $\rho < 0$ per architecture and fleet size (nominal runs). 38

5.9 E6: age of the cached DTC directive with the link up (dotted) and with a link failure from 90 to 120 s (solid). 39
5.10 MASooda Live Airspace on the IITH campus during the weather-cell scenario: drones colour-coded by type on road-aligned corridors with layer labels, a storm cell drawn as a restricted area over the Main Gate, vertiports with queue counts, and the scenario and architecture controls. 41
5.11 Corridor-benchmark dashboard at $t = 75$ s: the malicious agent and its falsified "ghost" report, the STL robustness trace, emergency progress against free flow, cumulative ADMM events and iterations, and the attack, link and violation timeline. 42

List of Tables

2.1 Coverage of the concerns of MASooda by the reviewed literature. "Formal" marks formal guarantees or formal specification monitoring. 10

3.1 Main notation used in the report. 12

4.1 Per-tick computational cost of the components. n_g, e_g: agents and pair constraints in a conflict group; L: layers; n_s: series cells per pack. 28

5.1 Base simulation configuration. 30
5.2 Week 6 experiment matrix. Run counts are those stored for each experiment folder; E5 adds only the combinations that E1–E4 lack. 31
5.3 Failure-mode test matrix. 32
5.4 E1: ADMM scaling (hierarchical, no attack, iteration limit 50; mean ± 95% CI over three seeds). 33
5.5 E2: merging, throughput and mean corridor utilisation (occupancy divided by the capacity of 15), hierarchical architecture. 34
5.6 E3: emergency priority (means over seeds 42–44): compliance rate, emergency delay against free flow, time saved against the control run, delay imposed on other agents, and travel times of the emergency agent (E) and of the same agent in the control run (C). Confidence intervals are wide (Section 5.2). 35
5.7 E4: architectures under a position-falsification attack (N = 60, five seeds; 50 m offset three seeds). *Dominated by the horizontal-metric artefact of Section 5.1.2. 36
5.8 Layer agreement on the IITH campus map (300 drones, 240 s, seed 42, hierarchical). 39
5.9 Headless results of the scripted scenarios on the IITH campus (seed 42). . . . 40
5.10 Steady-state outputs of the drone health model with default parameters at 35 °C and 542 m elevation (SoH 100%, energy counted down to 5% SoC, pack heat including the RC branch). 40
5.11 Status of the project goals. 43

A.1 Simulation, local behaviour, ADMM, DTC and resilience parameters. 55
A.2 Drone health model: shared parameters and thresholds. 56
A.3 Drone health model: parameters per vehicle type. 57

Acronyms

AAM Advanced Air Mobility. *(p. vi)*
ADMM Alternating Direction Method of Multipliers. *(p. vi)*

BMS Battery Management System. *(p. vi)*

ConOps Concept of Operations. *(p. vi)*
CPA Closest Point of Approach. *(p. vi)*

DTC Drone Traffic Controller (the global overseer). *(p. vi)*

eVTOL electric Vertical Take-Off and Landing aircraft. *(p. vi)*

GNSS Global Navigation Satellite System. *(p. vi)*

HDOP Horizontal Dilution of Precision. *(p. vi)*

MAS Multi-Agent System. *(p. vi)*
MPC Model Predictive Control. *(p. vi)*

OCV Open-Circuit Voltage. *(p. vi)*
OSM OpenStreetMap. *(p. vi)*

PHM Prognostics and Health Management. *(p. vi)*

RSSI Received Signal Strength Indicator. *(p. vi)*

SoC State of Charge. *(p. vi)*
SoH State of Health. *(p. vi)*
STL Signal Temporal Logic. *(p. vi)*

TFR Temporary Flight Restriction. *(p. vi)*

UAM Urban Air Mobility. *(p. vi)*

UAS Unmanned Aircraft System. *(p. vi)*

UTM Unmanned Aircraft System Traffic Management. *(p. vi)*

V2V Vehicle-to-Vehicle communication. *(p. vi)*

W-MSR Weighted Mean-Subsequence-Reduced algorithm. *(p. vi)*

1 Project Introduction

1.1 Vision & Motivation

Small drones and electric vertical take-off and landing (eVTOL) aircraft are moving from demonstrations into everyday urban services: medical deliveries between hospitals, parcel delivery from dark stores, inspection of infrastructure and, in the near future, passenger transport. Each of these services is attractive on its own, but together they create something cities have never had before: hundreds of autonomous aircraft sharing the same few hundred metres of airspace above streets, campuses and buildings. Conventional air traffic control, in which a human controller clears every movement, does not scale to this density. Regulators have therefore proposed Unmanned Aircraft System Traffic Management (UTM) as a service-based, highly automated layer for low-altitude airspace (Federal Aviation Administration, 2020; European Union Aviation Safety Agency, 2021; Ministry of Civil Aviation, Government of India, 2021a).
Designing such a traffic manager is a multi-agent problem. Every drone is an agent with its own mission, priority, sensors and energy budget. Agents interact locally (two drones approaching the same junction must agree on who yields) and globally (a corridor that is already full should not receive more traffic). Two extreme architectures are possible:

- A **centralised** controller that sees every aircraft and computes every command. It has the global picture, but its computation and communication grow with the fleet, it becomes a single point of failure, and it trusts whatever the aircraft report.
- A **fully decentralised** swarm in which each drone reacts only to the neighbours within its communication range. It is fast and has no single point of failure, but it cannot see congestion beyond its range, cannot enforce fleet-wide priorities and is exposed to neighbours that lie about their position.

MASooda (*Multi-Agent System for Hierarchical, Priority-Aware, and Resilient Urban Air Traffic Management*) takes the middle path. Fast, short-horizon decisions are taken locally by the drones themselves; slow, airspace-wide decisions are taken by a central Drone Traffic Controller (DTC). The DTC does not fly the drones: it allocates cor-

ridors, altitude layers and priorities, and the drones keep operating when the DTC is unreachable because they keep their last directive and continue to resolve conflicts among themselves.
Three further concerns shape the design:

1. **Heterogeneity and priority.** A medical drone carrying blood should not wait behind a parcel. Priorities must be respected both by the DTC (who gets which layer) and by the local negotiation (who yields in a conflict).
2. **Adversarial and faulty information.** Position reports can be spoofed, either by a malicious operator or by a GNSS fault. A traffic manager that blindly averages neighbour reports can be steered by a single liar.
3. **Verifiable safety.** Minimum separation must be monitored continuously and expressed as a formal specification, so that a violation is detected, quantified and logged rather than noticed afterwards.

During the project a fourth concern emerged as the most original direction. Every drone in the fleet is battery powered, and its state of charge, battery health and pack temperature decide how far it can safely fly. Existing traffic-management work treats vehicles as energy-agnostic, while battery prognostics work stops at the single vehicle (Chapter 2). MASooda therefore gives every drone a battery management system (BMS) model and lets health information take part in traffic decisions such as right of way for landing, diversion to a reachable pad and climb costs in layer selection.
Finally, the project is grounded in a real place. The final system flies drones along corridors traced from the IIT Hyderabad campus road map and from OpenStreetMap roads of Hyderabad, between named vertiports such as the campus hospital, the hostels and the messes, under the hot-climate conditions of the region.

1.2 Goals

The project goals were fixed in the team's six-week workflow and later extended. The **core goals** were:

1. Build a 2D/2.5D urban air-traffic simulation with about 50–80 heterogeneous agents of three types (emergency, eVTOL and delivery) and two to three pre-defined corridors.
2. Implement a local coordination tier based on Boids-style separation, alignment and cohesion, with corridor following and corridor merging.
3. Resolve short-horizon conflicts between agents with ADMM, including the general case of several agents meeting at a merge point.
4. Implement a centralised overseer (the DTC) for corridor allocation, macro routing and priority arbitration, and keep the system operating when the DTC link fails.
5. Model one adversarial scenario with falsified position information and mitigate it with a trusted-range consistency check on neighbour position reports, with W-MSR-style trimming in the consensus steps.

6. Monitor one STL specification for minimum separation live, through its quantitative robustness.
7. Compare the hierarchical architecture with purely centralised and purely decentralised (undefended) architectures under identical initial conditions.
8. Deliver reproducible experiments, plots, a live dashboard and a rehearsed demonstration.

The **extended goals**, pursued after the core was complete, were:

9. Constrain all flights to road-aligned corridors on real maps (IITH campus and Hyderabad), with vertiports, a geofence, corridor tubes and junction merging.
10. Use the full layered airspace: seven altitude layers, a direction-dependent (semicircular) layer rule, a reserved emergency layer, and V2V negotiation of layers with DTC congestion pricing.
11. Give every drone a physically motivated power, battery, BMS, thermal, link and navigation-quality model, with onboard responses and a DTC right-of-way rule for unhealthy drones.
12. Provide an operator console with real air-traffic-control style commands and a library of scripted, replayable scenarios.
13. Stream the simulation live to a browser with network-theory, ADMM, safety and health panels, and deploy it publicly.

1.3 Scope

The simulation represents each vehicle by its position, velocity, destination, priority and (in the extended system) its energy state. It is a kinematic 2.5D model: horizontal motion is continuous, while altitude is a discrete layer. The following were explicitly out of scope: true 3D aerodynamics and rendering, certification-grade safety, multiple simultaneous attack types, full epistemic-logic belief modelling and reinforcement-learning policies. Corridors in the map-constrained mode are a research planning backbone and are not approved flight paths.

1.4 Contributions of this Work

At the system level, this work makes the following contributions:

- A complete two-tier architecture in which Boids, priority-weighted consensus ADMM, a priority-aware DTC, a trusted-range report defence (with W-MSR trimming in the consensus steps) and an STL monitor run in one deterministic per-tick pipeline, with shared, frozen interfaces and a single configuration source.
- A fair comparison framework for hierarchical, centralised and decentralised architectures from bitwise-identical initial conditions, with a six-experiment, 141-run campaign and confidence intervals.

- A map-constrained live airspace on the IITH campus and Hyderabad road networks with corridor tubes, geofences, seven layers, tactical altitude shifts and a V2V layer-agreement protocol based on consensus dual ascent.
- A drone health model (momentum-theory power, equivalent-circuit battery, cell-level BMS, lumped thermal model, link and GNSS quality, fault injection) integrated with traffic decisions, together with an operator console and ten replayable scenarios.
- An honest account of measurement artefacts and open issues found by the team's own audit, so that every reported number can be interpreted correctly.

1.5 Organisation of the Report

Chapter 2 reviews the literature on UTM, flocking, distributed optimisation, resilient consensus, temporal-logic monitoring and battery health management, and identifies the research gap. Chapter 3 fixes the notation and the mathematical background. Chapter 4 formulates the problem and describes every component of the solution, with theoretical and complexity analysis. Chapter 5 describes the simulation setup, the experiments and the live demonstration, and discusses the results and limitations. Chapter 6 concludes and outlines future work. The appendices list the full parameter set and the scripted scenarios.

2 Literature Review

This chapter reviews the six bodies of work that MASooda draws on: concepts of operations for low-altitude traffic management, local coordination rules, distributed optimisation for conflict resolution, resilient consensus under adversarial information, formal runtime monitoring, and battery health management for electric aircraft. It ends with the research gap that motivates the health-aware extensions of the project.

2.1 Traffic Management Concepts for Low-Altitude Airspace

UTM and U-space. NASA's UTM research programme proposed a service-oriented ecosystem for small unmanned aircraft in which operators share flight intent and constraints with service suppliers instead of receiving clearances from a controller (Kopardekar et al., 2016). The FAA UTM Concept of Operations formalises this as a federated system of strategic deconfliction, constraint sharing and conformance monitoring, and leaves endurance and energy with the operator (Federal Aviation Administration, 2020). In Europe, the U-space regulation defines designated airspace in which service providers offer network identification, geo-awareness, flight authorisation and traffic information (European Union Aviation Safety Agency, 2021). The FAA Urban Air Mobility ConOps describes corridors as three-dimensional route segments with performance requirements (Federal Aviation Administration, 2023).

India. The Drone Rules 2021 divide Indian airspace into red, yellow and green zones that operators must consult through the Digital Sky platform (Ministry of Civil Aviation, Government of India, 2021b). The National UTM Policy Framework lists *UAS health monitoring*, the real-time monitoring of the health and status of unmanned aircraft, and *contingency management* as UTM services (Ministry of Civil Aviation, Government of India, 2021a), but does not specify how health information should change traffic decisions. This observation is central to the extended scope of MASooda.

Airspace structure. Two structural ideas from manned aviation are reused in MASooda. Separating traffic into discrete altitude layers turns a three-dimensional problem into a set of coupled two-dimensional problems. The semicircular (hemispherical)

cruising-level rule assigns levels by direction of flight, so that opposite-direction traffic is vertically separated by construction (International Civil Aviation Organization, 2005).

2.2 Local Coordination: Flocking, Platooning and Car-Following

Reynolds' Boids model showed that three local rules, *separation* from close neighbours, *alignment* with their velocity and *cohesion* towards their centre, are sufficient to produce coherent group motion without any central planner (Reynolds, 1987). Olfati-Saber gave a control-theoretic treatment of flocking with potential functions and proved conditions for collision-free cohesive motion (Olfati-Saber, 2006). Boids-style rules are cheap, need only neighbour information and degrade gracefully, which makes them a natural local tier. They do not, however, guarantee separation or encode priorities. For traffic in a lane, constant time-headway car-following laws, in which the desired speed is proportional to the gap to the leader minus a standstill distance, provide stable spacing and are the basis of vehicle platooning (Treiber et al., 2000). MASooda combines both: Boids terms for neighbours in general, and a time-headway law for the drone directly ahead in the same corridor.

2.3 Distributed Optimisation for Conflict Resolution

The Alternating Direction Method of Multipliers splits an optimisation problem into local subproblems that are coordinated through consensus variables and dual updates (Boyd et al., 2011). It suits multi-agent problems because each agent solves a small, often closed-form, subproblem and exchanges only a few variables with its neighbours. Ni, Pan and Gao used ADMM for robust multi-robot trajectory optimisation with collision and kinematic constraints (Ni et al., 2022). Because exact collision-avoidance constraints are non-convex, a common approach is to linearise them around the current configuration, giving half-space constraints that are conservative with respect to the true separation condition (Augugliaro et al., 2012). Priority in tactical deconfliction has been studied for U-space by Jover, Bermúdez and Casado, who extend a tactical algorithm with priority values from the operation class, so that lower-priority drones absorb most of the deviation (Jover et al., 2022). Zongo and Wei showed that the energy cost of conflict resolution is small on average but heavily right-skewed, with tails above 40% for some flights (Zongo et al., 2026). Learning-based separation assurance for heterogeneous fleets (Sharifi et al., 2026) and decentralised edge-AI conflict management (Agnihotri et al., 2025) are active alternatives, but neither models per-vehicle energy or health.

2.4 Hierarchical versus Centralised and Decentralised Control

Hierarchical multi-agent control separates timescales: a slow supervisory layer sets references, constraints or prices, and a fast local layer tracks them (Scattolini, 2009). This is the structure of MASooda: the DTC updates every two seconds, while local behaviour and ADMM run every 0.1 s tick. Hierarchy also gives a natural failure mode: when the supervisor is lost, the local layer continues with cached references. Price-based coordination, in which a central agent publishes congestion prices and agents respond with local decisions, is the dual-decomposition view of the same idea and underlies the layer-agreement protocol of Section 4.2.6.

2.5 Resilient Consensus under Adversarial Information

Plain consensus averages neighbour values, so a single neighbour that reports an arbitrary value can drive every honest agent to any value it chooses. LeBlanc et al. introduced the Weighted Mean-Subsequence-Reduced (W-MSR) algorithm, in which each agent discards up to f values larger than its own and up to f values smaller than its own before averaging (LeBlanc et al., 2013). They proved that honest agents reach consensus within the range of honest initial values if the communication graph is $2f$ 1-robust for an f-total adversary, or f $1, f$ 1-robust for an f-local adversary. Position spoofing of GNSS receivers is a realistic threat for small drones (Kerns et al., 2014). Physical checks, such as comparing a reported position with a locally sensed range, provide an additional trusted signal against which reports can be checked.

2.6 Formal Runtime Verification with Signal Temporal Logic

Signal Temporal Logic expresses properties of real-valued signals over time, such as "always keep at least 8 m from every other drone" (Maler et al., 2004). Its quantitative (robustness) semantics assigns each signal a real number whose sign says whether the property holds and whose magnitude says by how much (Fainekos et al., 2009; Donzé et al., 2010). Robustness can be monitored online over a sliding window and used both as a safety indicator and as an objective for control. Pant et al. used STL robustness to control fleets of drones with separation and mission specifications (Pant et al., 2018). In MASooda, STL is used as an online monitor; energy and thermal specifications are identified as future work.

2.7 Battery Health Management for Electric Aircraft

Battery prognostics and health management for electric UAVs is mature at the level of a single vehicle. Saha et al. built a battery health management system with particle-filter end-of-discharge prediction (Saha et al., 2011). Bole, Daigle and Gorospe estimated state

of charge with an unscented Kalman filter and detected parasitic loads in flight (Bole et al., 2014). Extended Kalman filtering is the classical approach to state-of-charge estimation in battery management systems (G. L. Plett, 2004). Kulkarni, Hogge and Quach flight-tested remaining-flight-time prediction (Kulkarni et al., 2018). Such estimates feed onboard decision-making frameworks that choose between nominal and contingency actions (Baculi et al., 2019; Quinones-Grueiro et al., 2020), real-time assessment of whether a delivery mission is still feasible (Taye et al., 2025), and health-aware operating policies for urban air mobility learned with deep reinforcement learning (Montazeri et al., 2024). Health-aware replanning (Darrah, 2023), prognostics-informed battery reconfiguration (Sharma et al., 2021) and state-of-health-aware path planning (Salinas-Camus et al., 2023) use these signals, but always for one vehicle. Yang et al. showed that eVTOL batteries face stricter power and energy requirements than electric-vehicle batteries (Yang et al., 2021), and Zhao et al. list multi-fault diagnosis and cybersecurity as open problems for drone battery systems (Zhao et al., 2025). The modelling tools used in MASooda are standard: momentum theory for rotor power (Leishman, 2006), equivalent-circuit battery models (Gregory L. Plett, 2015), a lumped energy balance for pack temperature (Bernardi et al., 1985) and log-distance path loss for the command link (Rappaport, 2002).

2.8 Research Gap

Table 2.1 summarises how the reviewed work covers the concerns of MASooda. Combining these observations, the team identified the following gaps, which motivate the extended scope of the project:

G1 The traffic manager is health-blind. Priority, layer allocation and deconfliction weights come from operation class and geometry (Jover et al., 2022; Sharifi et al., 2026; Ni et al., 2022). Live battery state does not decide who yields, who gets which layer or who lands first.

G2 Health management stops at the single vehicle. Prognostics produce exactly the signals a traffic manager needs, but decisions to abort, divert or swap a pack are taken onboard without knowledge of surrounding traffic or shared landing capacity (Saha et al., 2011; Darrah, 2023; Sharma et al., 2021; Baculi et al., 2019; Quinones-Grueiro et al., 2020; Taye et al., 2025).

G3 The energy cost of deconfliction is measured but not managed. Its tail can be large (Zongo et al., 2026), yet it is assigned by geometry and fixed priority, so a drone with little reserve may be the one that manoeuvres.

G4 Health telemetry is an unprotected input. Once health affects priority, a falsified health report becomes an attack on the traffic manager. Resilient consensus (LeBlanc et al., 2013) has not been combined with health-aware prioritisation in the work the team found.

G5 Safety specifications ignore energy and heat. STL monitoring covers sepa-

ration and missions (Pant et al., 2018); energy-reserve and thermal specifications are missing from multi-drone traffic management.

G6 Hot-climate operations are not studied. Indian policy asks for real-time health monitoring (Ministry of Civil Aviation, Government of India, 2021a), but no traffic-management study evaluates heat-wave conditions such as the 40–44 °C summers of Hyderabad.

Work	Multi-agent	Priority	Adversary	Formal	Health
Boids and flocking (Reynolds, 1987; Olfati-Saber, 2006)	yes	–	–	–	–
ADMM trajectories (Ni et al., 2022)	yes	–	–	–	–
Priority-aware U-space (Jover et al., 2022)	yes	yes	–	–	–
W-MSR consensus (LeBlanc et al., 2013)	yes	–	yes	proofs	–
Fly-by-Logic (Pant et al., 2018)	yes	–	–	STL	–
Energy overhead (Zongo et al., 2026)	yes	–	–	–	energy
Battery PHM (Saha et al., 2011; Bole et al., 2014)	–	–	–	–	yes
Health-aware replanning (Darrah, 2023)	–	–	–	–	yes
MASooda (this work)	yes	yes	yes	STL	yes

Table 2.1: *Coverage of the concerns of MASooda by the reviewed literature. "Formal" marks formal guarantees or formal specification monitoring.*

The core MASooda system addresses the multi-agent, priority, adversarial and formal-monitoring columns together. The health model, the DTC right-of-way rule for unhealthy drones, the battery-weighted climb cost in layer agreement and the heat-wave scenario are first steps towards G1, G2 and G6. G3, G4 and G5 are planned extensions (Section 6.2).

3
Notations & Preliminaries

3.1 Notations

3.1.1 Agents, Time and Airspace

The fleet is a finite set of agents $\mathcal{V} = \{1, \dots, N\}$. Time advances in ticks $t_k = k\,\Delta t$ with $\Delta t = 0.1\,\mathrm{s}$. Each agent i has a horizontal position $\mathbf{p}_i \in \mathbb{R}^2$, a horizontal velocity $\mathbf{v}_i \in \mathbb{R}^2$, a discrete altitude layer $\ell_i \in \{0, \dots, L-1\}$, a type $\tau_i \in \{\textsc{Emergency}, \textsc{eVTOL}, \textsc{Delivery}\}$ and an integer priority P_i, where a *smaller* number means a *higher* priority. Table 3.1 lists the symbols used throughout the report.

3.1.2 Norms, Sets and Operators

$\|\cdot\|$ is the Euclidean norm, $\langle\cdot,\cdot\rangle$ the inner product and $x = \max 0, x$. $\Pi_{\mathcal{C}}\cdot$ denotes Euclidean projection onto a closed convex set $\mathcal{C}$. $\mathbb{1}\cdot$ is the indicator of a condition. $\mathbf{G}$ and $\mathbf{F}$ are the STL "globally" (always) and "finally" (eventually) operators. Confidence intervals are 95% two-sided intervals of the mean based on the t distribution over seeds.

3.2 Background on Preliminaries

3.2.1 2.5D Kinematics

Each agent is a horizontal double integrator with clamped acceleration and speed,

$$\mathbf{a}_i t_k = \mathrm{clip}_{a_{\max}}\left(\frac{\mathbf{v}_i^{\mathrm{des}} t_k - \mathbf{v}_i t_k}{\tau_v}\right), \tag{3.1}$$

$$\mathbf{p}_i t_{k1} = \mathbf{p}_i t_k \;\; \mathbf{v}_i t_k \, \Delta t, \tag{3.2}$$

$$\mathbf{v}_i t_{k1} = \mathrm{clip}_{v_{\max}}\left(\mathbf{v}_i t_k \;\; \mathbf{a}_i t_k \, \Delta t\right), \tag{3.3}$$

where $\mathrm{clip}_c \mathbf{y} = \mathbf{y}\ \min 1, c/\|\mathbf{y}\|$, $\mathbf{v}_i^{\mathrm{des}}$ is the desired velocity produced by the higher layers and $\tau_v = 0.5\,\mathrm{s}$ is a velocity-tracking time constant. The vertical coordinate is not in-

Symbol	Meaning
N, $\mathcal{V}$	number of agents, set of agents
Δt	simulation time step (0.1 s)
$\mathbf{p}_i$, $\mathbf{v}_i$, $\mathbf{a}_i$	horizontal position, velocity and acceleration of agent i
ℓ_i, L, H	altitude layer of agent i, number of layers, layer spacing (15 m)
P_i	priority of agent i (0 emergency, 1 eVTOL, 2 delivery)
$v_{\max}$, $a_{\max}$	speed limit (15 m/s) and acceleration limit (4 m/s^2)
v_c	cruise speed (12 m/s nominal, scaled by mission type)
$d_{\min}$	minimum separation (8 m)
$d_{ij}t$	distance between agents i and j at time t
$\mathcal{G}_t = \mathcal{V}, \mathcal{E}_t$	communication (V2V) graph at time t
$\mathcal{N}_i$	neighbours of agent i in $\mathcal{G}_t$
r_c, k	communication radius (40 m) or number of nearest neighbours (6)
ρt	STL robustness of the separation specification at time t
ρ_{admm}	ADMM penalty parameter
$\mathbf{x}_i$, $\mathbf{z}$, $\mathbf{u}$	ADMM local variables, consensus copies and scaled duals
w_i	priority weight of agent i in ADMM
f	number of extreme values trimmed on each side by W-MSR
C	capacity of one corridor (or sector) on one layer (15)
$\lambda_{s,\ell}$	congestion price of layer ℓ in sector s
SoC, SoH	state of charge and state of health of a battery
T_{pack}, T_{mot}, T_{amb}	pack, motor and ambient temperatures

Table 3.1: *Main notation used in the report.*

tegrated: it is the discrete layer ℓ_i, nominally at height 30 m $\ell_i H$ above ground with $H = 15$ m (layers L0–L6 at 30–120 m). Changing layer is an instantaneous discrete event; in the map-constrained mode it is guarded by a safety check (Section 4.2.5), while the corridor-mode engine used for the experiment campaign applies a DTC layer change directly. This "2.5D" abstraction keeps the computational model simple while capturing vertical separation, which is the dominant separation mechanism in structured airspace.

3.2.2 Communication Graphs and Network Measures

Agents exchange position reports over a V2V graph built every tick either by radius, $\mathcal{E}_t = \{i, j : \|\mathbf{p}_i - \mathbf{p}_j\| \leq r_c\}$, or by k nearest neighbours (made undirected). The live system reports standard network measures of this graph: the degree deg$i = |\mathcal{N}_i|$, the density $2|\mathcal{E}|NN - 1$, the transitivity (global clustering coefficient, three times the number of triangles divided by the number of connected triples), the maximum degree centrality $\max_i$ deg$iN - 1$, the number of connected components and the share of the giant component, and the algebraic connectivity $\lambda_2\mathbf{L}$, the second-smallest eigenvalue of the graph Laplacian $\mathbf{L} = \mathbf{D} - \mathbf{A}$. The graph is connected if and only if $\lambda_2 > 0$, and larger λ_2 means faster consensus (Fiedler, 1973; Olfati-Saber et al., 2007). The DTC is drawn as an extra hub node connected to every airborne drone whose directive link is up.

3.2.3 Boids Rules

For agent i with neighbour set $\mathcal{N}_i$ and the subset $\mathcal{S}_i \subseteq \mathcal{N}_i$ of neighbours in the same corridor, the three Boids terms used in this work are

$$\mathbf{v}_i^{\text{sep}} = \sum_{j \in \mathcal{N}_i,\ 0<d_{ij}<r_{\text{sep}}} \frac{\mathbf{p}_i - \mathbf{p}_j}{d_{ij}^2}, \tag{3.4}$$

$$\mathbf{v}_i^{\text{ali}} = \frac{1}{|\mathcal{S}_i|} \sum_{j \in \mathcal{S}_i} \mathbf{v}_j - \mathbf{v}_i, \tag{3.5}$$

$$\mathbf{v}_i^{\text{coh}} = \left(\frac{1}{|\mathcal{S}_i|} \sum_{j \in \mathcal{S}_i} \mathbf{p}_j - \mathbf{p}_i \right) \cdot \mathbf{e}_{\parallel}\, \mathbf{e}_{\parallel}, \tag{3.6}$$

where $r_{\text{sep}} = 20\,\text{m}$ and $\mathbf{e}_{\parallel}$ is the unit vector along the corridor. The inverse-square separation term grows quickly as two drones approach. Cohesion acts only along the corridor and is switched off when any same-corridor neighbour is closer than r_{sep}, so that it never pulls drones into each other.

3.2.4 The Alternating Direction Method of Multipliers

ADMM solves problems of the form $\min_{\mathbf{x},\mathbf{z}} f\mathbf{x}\ \ g\mathbf{z}$ subject to $\mathbf{x} = \mathbf{z}$. With penalty $\rho_{\text{admm}} > 0$ and scaled dual variable $\mathbf{u}$, the iterations are (Boyd et al., 2011)

$$\mathbf{x}^{m1} = \arg\min_{\mathbf{x}}\ \ f\mathbf{x}\ \ \tfrac{\rho_{\text{admm}}}{2}\|\mathbf{x} - \mathbf{z}^m\ \ \mathbf{u}^m\|^2, \tag{3.7}$$

$$\mathbf{z}^{m1} = \arg\min_{\mathbf{z}}\ \ g\mathbf{z}\ \ \tfrac{\rho_{\text{admm}}}{2}\|\mathbf{x}^{m1} - \mathbf{z}\ \ \mathbf{u}^m\|^2, \tag{3.8}$$

$$\mathbf{u}^{m1} = \mathbf{u}^m\ \ \mathbf{x}^{m1} - \mathbf{z}^{m1}. \tag{3.9}$$

When g is the indicator of a convex set, the $\mathbf{z}$-update is a projection onto that set. Progress is measured by the primal residual $r^m = \|\mathbf{x}^m - \mathbf{z}^m\|$ and the dual residual $s^m = \rho_{\text{admm}}\|\mathbf{z}^m - \mathbf{z}^{m-1}\|$. The method stops when

$$r^m \le \sqrt{n}\,\epsilon_{\text{abs}}\ \ \epsilon_{\text{rel}} \max\|\mathbf{x}^m\|, \|\mathbf{z}^m\|, \qquad s^m \le \sqrt{n}\,\epsilon_{\text{abs}}\ \ \epsilon_{\text{rel}}\,\rho_{\text{admm}}\|\mathbf{u}^m\|, \tag{3.10}$$

with n the number of consensus entries. For closed, proper, convex f and g and a problem that has a saddle point, the residuals converge to zero and the objective converges to the optimum.

3.2.5 Resilient Consensus and W-MSR

In the W-MSR update, agent i with own value x_i receives values $\{x_j\}_{j \in \mathcal{N}_i}$. It sorts them, removes the f largest values that are greater than x_i (or all of them if there are fewer than f) and the f smallest values that are less than x_i, and averages its own value with the remaining set $\mathcal{R}_i$:

$$x_i = \frac{1}{1\ \ |\mathcal{R}_i|} \left(x_i\ \ \sum_{j \in \mathcal{R}_i} x_j \right). \tag{3.11}$$

A graph is *r-robust* if, for every pair of non-empty disjoint vertex subsets, at least one subset contains a vertex with at least r neighbours outside its subset. LeBlanc et al. prove that under an f-local adversary model, W-MSR achieves resilient asymptotic consensus if the graph is $2f$ 1-robust (LeBlanc et al., 2013). Vector-valued reports are filtered one coordinate at a time.

3.2.6 Signal Temporal Logic and Robustness

An STL formula φ is interpreted over a signal $\mathbf{s} : 0, T \to \mathbb{R}^n$. For a predicate $\mu\mathbf{s} \geq 0$, the robustness is $\rho\mu, \mathbf{s}, t = \mu\mathbf{s}t$. For the "always" operator over a window,

$$\rho\mathbf{G}_{a,b}\varphi, \mathbf{s}, t = \min_{t' \in ta,\, tb} \rho\varphi, \mathbf{s}, t'. \tag{3.12}$$

The robustness is *sound*: $\rho > 0$ implies that the signal satisfies φ, and $\rho < 0$ implies that it violates φ (Fainekos et al., 2009; Donzé et al., 2010). Any perturbation of the signal smaller than its magnitude cannot change the verdict, and here this magnitude has a direct physical meaning in metres.

3.2.7 Rotor Power and Battery Equivalent Circuit

For a multirotor of total mass m with total rotor disc area A in air of density ρ_a, momentum theory gives the hover induced velocity $v_h = \sqrt{mg2\rho_a A}$ and, in level flight at airspeed v, the induced velocity

$$v_i v = \sqrt{-\tfrac{v^2}{2} \sqrt{\tfrac{v^4}{4} v_h^4}} \qquad \text{(Leishman, 2006)}. \tag{3.13}$$

An internal-resistance (Rint) equivalent circuit models a battery pack as an open-circuit voltage source OCVSoC in series with a resistance R (Gregory L. Plett, 2015). For an electrical power demand P, the pack current solves $P = \mathrm{OCV} - IR\,I$, which has the physical root

$$I = \frac{\mathrm{OCV} - \sqrt{\mathrm{OCV}^2 - 4RP}}{2R}, \tag{3.14}$$

and has no real solution when $\mathrm{OCV}^2 < 4RP$, i.e. when the pack cannot deliver the requested power. The simulated pack extends this circuit with one RC branch (a 1-RC Thevenin model with $R_1 = 0.5\,R_0$ and $R_1C_1 = 20\,\mathrm{s}$); Eq. (3.14) is then applied with $\mathrm{OCV} - V_{\mathrm{RC}}$ in place of OCV and the series resistance R_0 in place of R, while the plain Rint form is used in the energy-feasibility check.

4 Problem Formulation

4.1 Problem Statement

Consider N autonomous aircraft of three types sharing a bounded urban airspace that is structured into corridors and L discrete altitude layers. Each aircraft i starts at an origin, must reach a destination, has a priority P_i fixed by its mission (raised by the DTC for unhealthy aircraft), and observes only the aircraft within its communication range. A central DTC observes a coarse picture of the whole airspace through a link that may fail. Some aircraft may broadcast falsified positions. The problem is to design local and global decision rules such that, under identical initial conditions:

(S) Safety. The minimum-separation specification

$$\varphi_{\text{safe}} = \mathbf{G}\left(\min_{i \neq j} d_{ij} t \;\geq\; d_{\min}\right) \tag{4.1}$$

holds as often as possible, every violation is detected online and its depth is measured by the robustness ρt.

(L) Liveness and efficiency. Aircraft reach their destinations; throughput (arrivals per second) is high and delay with respect to free-flow travel is low.

(P) Priority. Higher-priority aircraft are preferred in corridor and layer allocation and in local conflicts; an emergency aircraft is delayed as little as possible.

(R) Resilience. A bounded number of aircraft reporting false positions cannot steer honest aircraft arbitrarily, and the loss of the DTC link degrades performance gracefully rather than causing failure.

(H) Health awareness (extended scope). No aircraft runs out of energy in flight, and unhealthy aircraft are given the right of way to land at a reachable pad.

The problem is solved in two execution phases, which share the core decision modules (Boids separation and alignment, ADMM, the DTC overseer, the report defence and the STL monitor) but differ in their world model.

4.1.1 Execution Phase #1: Corridor Benchmark

The first phase, carried out over the six-week workflow, uses an abstract world designed for controlled experiments: three parallel straight corridors of length 1500 m and width 20 m (the width is used to count corridor occupancy; agents are pulled towards the centreline but not confined to it), with centrelines 50 m apart, an entry zone ($x \in 0, 100$ m), a merge zone ($x \in 700, 800$ m), from whose start onward an agent follows a DTC corridor reassignment, and an exit zone ($x \in 1400, 1500$ m). Each corridor has a nominal capacity of 15 agents per altitude layer. The fleet of 30–75 agents is drawn from a seeded type mix of 10% emergency, 30% eVTOL and 60% delivery, and starts in the entry zone or, when it is too crowded, just beyond it ($x > 100$ m), placed by rejection sampling with $d_{\min}$ spacing; the sampling window grows until every agent fits. Figure 4.1 shows a snapshot of this world. Its purpose is to compare architectures, densities and attacks under exactly controlled conditions.

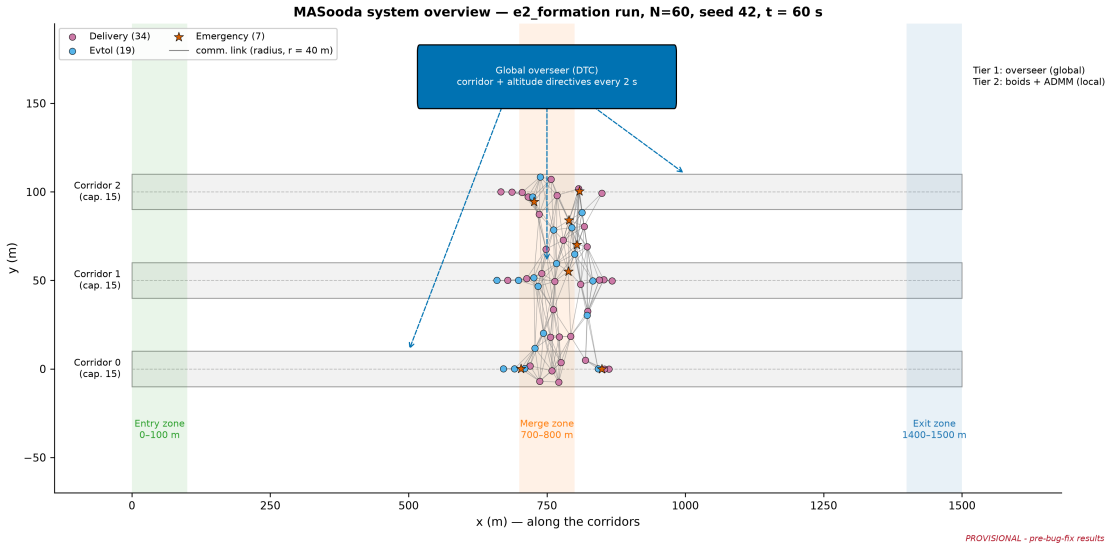


Figure 4.1: *Corridor benchmark world (60 agents, seed 42, $t = 60$ s): three corridors with entry, merge and exit zones, V2V links of radius 40 m, and the DTC sending corridor and altitude directives every 2 s.*

4.1.2 Execution Phase #2: Map-Constrained Live Airspace

The second phase replaces the abstract corridors by road-aligned corridors on real maps, because urban drones are expected to follow existing transport corridors, which concentrate both demand and conflicts at junctions.

- **IITH campus (default).** Roads, the campus perimeter as a polygon geofence and 13 named vertiports (Hospital, Main Gate, Guest House, Convention Centre, LHC, Admin Block, Academic Area, Mess A and B, Old and New Hostels, SNCC and Sampoorna) were traced by hand from the campus map (Figure 4.2) at about 0.95 m per pixel. The resulting corridor graph has 105 junctions.

- **Hyderabad roads.** Major OpenStreetMap road classes (motorway to tertiary) in a 3.1 km × 2.2 km area of western Hyderabad (centred at 17.44° N, 78.365° E) are projected into a local tangent-plane frame in metres, snapped into a routable graph and reduced to the largest connected component (1533 junctions). Eight vertiports are placed by deterministic farthest-point sampling and set back about 45 m from the road, each with a short approach leg.

Every flight follows the Dijkstra shortest path along the corridor graph between its origin and destination vertiports. Emergency flights go to hospitals and deliveries leave depots. The airspace has seven layers ($L = 7$, 15 m apart): eastbound traffic uses the even layers $\{0, 2, 4\}$, westbound traffic the odd layers $\{1, 3, 5\}$, and layer 6 is reserved for emergencies. Two boundary conditions are enforced after every dynamics step: a drone more than half a corridor width from its route centreline is projected back onto the tube wall and its outward velocity is removed (a sliding wall), and a drone outside the geofence is projected back just inside it and stopped (on the Hyderabad map, which has no perimeter polygon, the geofence is the rectangular map extent). Departures wait on the pad until the junction where the approach leg joins the corridor is clear on the departure layer, and a final approach descends vertically within 10 m of the pad.

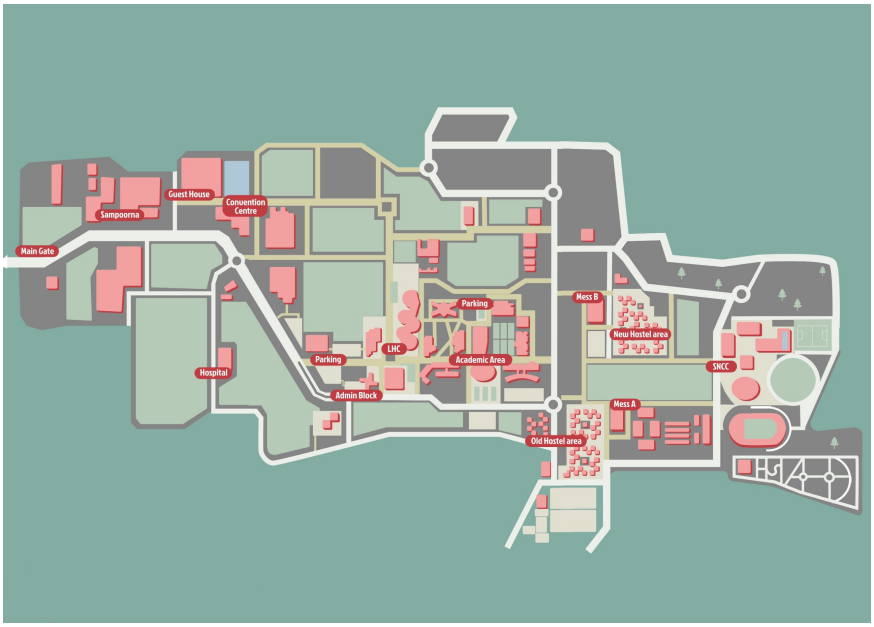


Figure 4.2: *IIT Hyderabad campus map used to trace the corridor network, the geofence and the 13 vertiports of the live airspace.*

4.2 Solution Approach

4.2.1 Architecture Overview

MASooda is organised in two tiers and two cross-cutting layers (Figure 4.3). All modules exchange data only through a shared data model (agent state, communication graph,

overseer directive, conflict event, attack state, robustness record and run metrics) and read every parameter from one configuration file. The data model was frozen at the end of Week 2 and the module interfaces by Week 4, after which changes required agreement from the owner and every consumer. Each module first shipped a mock (for example a hand-written list of agents or a fixed directive) so that no team member was ever blocked by another.

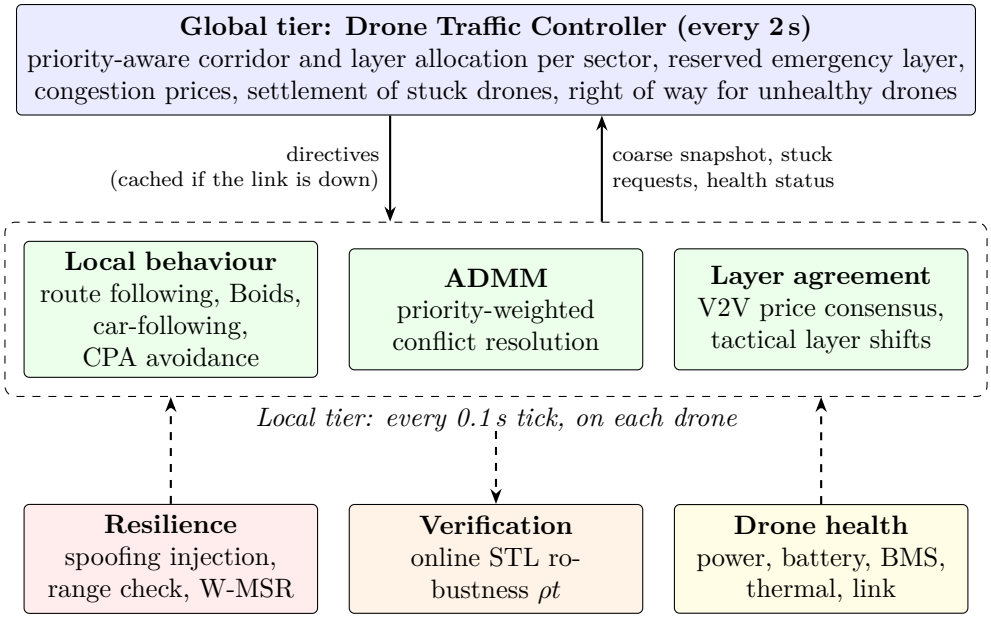


Figure 4.3: *The MASooda architecture. The DTC sets slow, airspace-wide references; the drones resolve fast, local interactions among themselves. Filtered reports from the resilience layer and health states feed the local tier, and the verification layer observes the outcome independently.*

4.2.2 Per-Tick Pipeline

Every tick executes the same deterministic sequence (Figure 4.4). Operator and scripted commands are applied first, so that a run can be replayed exactly from its command log. Departures are released when their pad and junction are clear. On DTC ticks, the DTC computes directives through its possibly failing link. The V2V graph is rebuilt and the attack (if any) falsifies broadcast positions. Layer negotiation runs once per second. Each receiver then filters the reports it hears, and local behaviour turns the filtered neighbour picture into a desired velocity, ADMM replaces the desired velocities of conflicting groups with negotiated ones, tactical altitude shifts handle imminent in-plane conflicts, and finally dynamics and boundary conditions advance the state, the STL monitor evaluates it and the health model is updated.

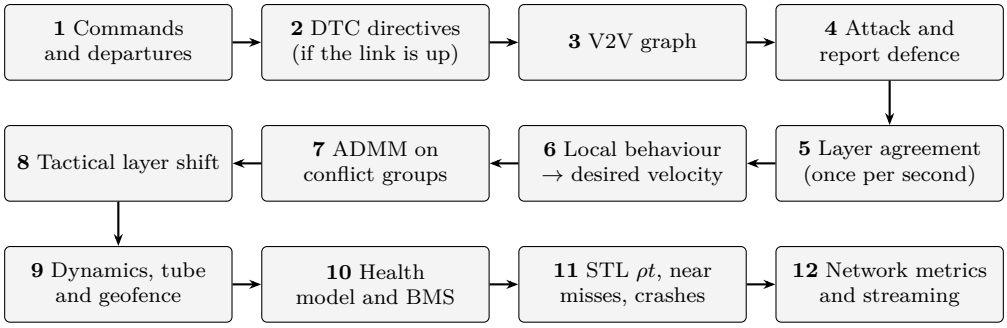


Figure 4.4: *Order of operations within one 0.1 s tick of the map-constrained engine. The corridor benchmark uses the same order without the map-specific steps.*

4.2.3 Local Tier: Corridor Following, Boids and Car-Following

Corridor benchmark. The desired velocity of agent i is a weighted sum of a goal term and the Boids terms of Chapter 3,

$$\mathbf{v}_i^{\text{des}} = \text{clip}_{v_{\max}}\Big(w_g\,\mathbf{v}_i^{\text{goal}}\;\; w_{\text{sep}}\,\mathbf{v}_i^{\text{sep}}\;\; w_{\text{ali}}\,\mathbf{v}_i^{\text{ali}}\;\; w_{\text{coh}}\,\mathbf{v}_i^{\text{coh}}\Big)\,, \tag{4.2}$$

with $w_g = 1$, $w_{\text{sep}} = 35$, $w_{\text{ali}} = 0.4$ and $w_{\text{coh}} = 0.1$. The goal term follows the centreline $y_c^\star = c \cdot 50\,\text{m}$ of the active corridor c through a look-ahead point $\mathbf{q} = x_i \;\; \ell_a,\;\; y_c^\star$ with $\ell_a = 40\,\text{m}$, plus a lateral correction:

$$\mathbf{v}_i^{\text{goal}} = v_c\,\frac{\mathbf{q} - \mathbf{p}_i}{\|\mathbf{q} - \mathbf{p}_i\|} - k_{\text{lat}}\,y_i - y_c^\star\,\mathbf{e}_y, \qquad k_{\text{lat}} = 0.5\ \text{s}^{-1}. \tag{4.3}$$

The target corridor comes from the DTC directive. If no directive is available the agent uses its last cached target, and otherwise its initial assignment. An agent changes corridor only inside or after the merge zone, and only when the target corridor has a free longitudinal gap of at least $g_m = 15\,\text{m}$ around it. If the gap is free, the lateral speed towards the target is capped at $2\,\text{m/s}$; if not, the agent stays in its lane and slows its forward speed by the factor $1 - k_v$ with $k_v = 0.5$ to open a gap. The weight w_{sep} was tuned in Week 3 to keep pairwise separation above 8 m during multi-agent merges in the validation scenarios.

Map-constrained mode. Each route is a polyline parametrised by arc length s. Agent i is projected onto its route to obtain its progress s_i, the tangent $\mathbf{t}_i$ and the left normal $\mathbf{n}_i$. The reference speed is the mission cruise speed (1.2, 1.0 and 0.9 times v_c for emergency, eVTOL and delivery), reduced by a braking profile near the pad, by any operator speed limit, and by an overheating limit from the health model. A constant time-headway car-following law reduces it further for every same-direction neighbour ahead in the same tube:

$$v_i^{\text{ref}} \leftarrow \min\left(v_i^{\text{ref}},\;\left[\frac{g_{ij} - d_0}{h}\right]\right), \qquad d_0 = 12\ \text{m},\;\; h = 1\ \text{s}, \tag{4.4}$$

where g_{ij} is the along-track gap. Pure-pursuit route following aims at a look-ahead point offset to the left of the centreline by a quarter of the corridor width (keep-left lanes, as on Indian roads), so that opposing flows are laterally separated. Tactical closest-point-

of-approach (CPA) avoidance handles crossing and head-on traffic: for each neighbour with relative position $\mathbf{r}$ and relative velocity $\mathbf{v}_r$,

$$t_{\text{cpa}} = -\frac{\langle \mathbf{r}, \mathbf{v}_r \rangle}{\|\mathbf{v}_r\|^2}, \qquad m = \|\mathbf{r} \;\; \mathbf{v}_r \, t_{\text{cpa}}\|, \tag{4.5}$$

and if $0 < t_{\text{cpa}} < 8\,\text{s}$ and the predicted miss distance m is below $1.6\,d_{\min}$, an avoidance velocity of magnitude $\min(6,\ 21.6\,d_{\min} - m \max t_{\text{cpa}}, 0.5)$ pushes away from the predicted CPA, and the drone also slows down if the threat is ahead. The final desired velocity adds Boids separation, alignment with same-direction drones and along-track cohesion (switched off whenever a segment mate is closer than $\max r_{\text{sep}},\ d_0\, h\|\mathbf{v}_i\|$, so that cohesion never pulls a drone inside its headway). Near junctions, a merge rule slows a joining drone by $1 - k_v$ until there is a gap on its layer, and platoons (leader within 40 m) are tracked for metrics.

4.2.4 Local Tier: Priority-Weighted Consensus ADMM

Conflict detection. Two airborne agents are in conflict when their horizontal distance is below the detection radius $r_d = 20\,\text{m}$ and they are on the same layer. In the corridor benchmark this is further restricted to agents inside the merge zone; in map mode it applies everywhere. Conflicts form an undirected graph, and each connected component is solved as one ADMM problem. An earlier version split large components into chunks of at most five agents, which could drop the constraint between two conflicting agents at a chunk boundary. This was corrected in Week 6 so that every detected conflict edge is present in exactly one solve.

Optimisation problem. For a conflict group $\mathcal{C}$, each agent i wants to reach its intended next position $\mathbf{p}_i = \mathbf{p}_i \;\; \mathbf{v}_i^{\text{des}} \Delta t$. ADMM finds the closest feasible set of next positions:

$$\min_{\{\mathbf{x}_i\}} \;\; \sum_{i \in \mathcal{C}} \frac{w_i}{2} \, \|\mathbf{x}_i - \mathbf{p}_i\|^2 \tag{4.6}$$

$$\text{s.t.} \;\; \langle \mathbf{n}_{ij},\ \mathbf{x}_i - \mathbf{x}_j \rangle \geq d_{\text{safe}} \quad \forall i, j \in \mathcal{P}, \tag{4.7}$$

$$\|\mathbf{x}_i - \mathbf{p}_i\| \leq v_{\max} \Delta t \quad \forall i \in \mathcal{C}, \tag{4.8}$$

where $\mathbf{n}_{ij} = \mathbf{p}_i - \mathbf{p}_j \|\mathbf{p}_i - \mathbf{p}_j\|$ is the unit vector from j to i at the current positions, $d_{\text{safe}} = 1.01\,d_{\min}$ includes a 1% margin, and $\mathcal{P}$ contains the same-layer pairs whose current distance is at most $d_{\text{safe}}\, 2 v_{\max} \Delta t$ (pairs farther apart cannot collide within one tick). The weights w_i encode priority: $w = 10$ for priority 0 (emergency, and also unhealthy drones in map mode), $w = 3$ for eVTOL and $w = 1$ for delivery. A large weight makes deviation expensive, so lower-priority agents absorb most of the manoeuvre. The resolved velocity is $\mathbf{x}_i - \mathbf{p}_i \Delta t$.

Consensus splitting. Each agent keeps one local variable $\mathbf{x}_i$. Every constraint in which it appears gets its own copy: a pair constraint $e = i, j$ owns copies $\mathbf{z}_{e,i}, \mathbf{z}_{e,j}$, and each agent's speed limit owns a copy $\mathbf{z}_{b,i}$. With K_i the number of copies of agent i (one

ball copy plus one per pair) and scaled duals $\mathbf{u}$, the updates are:

$$\mathbf{x}_i \leftarrow \frac{w_i\,\mathbf{p}_i \;\; \rho_{\text{admm}} \sum_{k \ni i} \mathbf{z}_{k,i} - \mathbf{u}_{k,i}}{w_i \;\; \rho_{\text{admm}} K_i}, \tag{4.9}$$

$$\mathbf{z}_{e,i}, \mathbf{z}_{e,j} \leftarrow \Pi_{\mathcal{H}_e}\big(\mathbf{x}_i \;\; \mathbf{u}_{e,i}, \;\; \mathbf{x}_j \;\; \mathbf{u}_{e,j}\big), \qquad \mathbf{z}_{b,i} \leftarrow \Pi_{\mathcal{B}_i} \mathbf{x}_i \;\; \mathbf{u}_{b,i}, \tag{4.10}$$

$$\mathbf{u}_{k,i} \leftarrow \mathbf{u}_{k,i} \;\; \mathbf{x}_i - \mathbf{z}_{k,i}. \tag{4.11}$$

The $\mathbf{x}$-update is a closed-form weighted average that every agent can compute in parallel from its own copies. The projection onto the half-space $\mathcal{H}_e$ splits the violation $\delta = d_{\text{safe}} - \langle \mathbf{n}_{ij}, \mathbf{a} - \mathbf{b} \rangle$ equally, moving $\mathbf{a}$ by $\frac{\delta}{2}\mathbf{n}_{ij}$ and $\mathbf{b}$ by $-\frac{\delta}{2}\mathbf{n}_{ij}$ when $\delta > 0$. The projection onto the speed ball $\mathcal{B}_i$ rescales the displacement to radius $v_{\max}\Delta t$. Iteration stops by the residual test (3.10) with $\epsilon_{\text{abs}} = 10^{-3}$, $\epsilon_{\text{rel}} = 10^{-4}$ and $\rho_{\text{admm}} = 1$, or after 50 iterations.

Non-convergence and fallback. If ADMM does not converge within the iteration limit, the group's original intents are returned and the event is marked as non-converged. The agents then rely on the Boids separation, CPA avoidance and tactical layer shifts for that tick. This is deliberate: a non-converged iterate is not guaranteed to satisfy the constraints and must not be presented as a resolution.

Altitude. Pairs on different layers produce no constraint, so ADMM solves independent planar problems per layer. When an in-plane conflict cannot be separated, the tactical layer shift of Section 4.2.5 moves the lower-priority drone vertically.

4.2.5 Global Tier: Drone Traffic Controller

Information. The DTC receives a coarse snapshot every $T_{\text{DTC}} = 2\,\text{s}$: the time, the occupancy of each corridor (or, in map mode, of each sector of a 3×2 grid over the map), and the identifier, type, priority and direction of each visible agent. It does not receive exact positions. Non-cooperative drones (no Remote-ID-style broadcast) and quarantined drones are invisible or ignored.

Priority-aware allocation. Algorithm 1 (Figure 4.5) assigns every visible agent a corridor and a layer (in map mode the sector is the drone's current sector and only the layer is allocated). Agents are processed in priority order (emergency, eVTOL, delivery; ties by identifier). Capacity $C = 15$ is counted per (corridor, layer). Emergencies always receive the reserved emergency layer. Other agents keep their previous slot if it still has capacity (stickiness avoids needless re-layering), and otherwise take the lowest allowed layer with space, where "allowed" means not the emergency layer and, when the direction is known, the correct parity under the semicircular rule. If every allowed slot is full, the least-loaded one is used and an overflow is counted.

Safety gate. In map mode, a directive is applied only if the target layer is locally clear: no drone on that layer within $3d_{\min}$, and no predicted CPA on that layer below $2d_{\min}$ within 10 s. Otherwise the change is deferred to the next DTC update and logged. Operator-pinned layers override the directive, and unhealthy drones keep their layer (a climb costs energy they may not have).

Tactical altitude shift. When the CPA check predicts a miss below $d_{\min}$ within 3 s,

Input: snapshot with agents $\mathcal{A}$, priorities P, types, directions; previous slots; capacity C; emergency layer ℓ_e.
Output: slot c_i, ℓ_i for every $i \in \mathcal{A}$.

1. Set every occupancy $oc, \ell \leftarrow 0$.
2. For each $i \in \mathcal{A}$ in increasing order of P_i, i:
   (a) If i is an emergency: $\ell \leftarrow \ell_e$; keep the previous corridor if $oc_i^{\text{prev}}, \ell_e < C$, else take $\arg\min_c oc, \ell_e$.
   (b) Else: let $\mathcal{L}_i$ be the allowed layers (not ℓ_e, correct parity). Candidates are the previous slot (if allowed), then for each $\ell \in \mathcal{L}_i$ in increasing order the emptiest corridor on ℓ. Take the first candidate with $o < C$; if none, take the least-loaded candidate and count one overflow.
   (c) $oc_i, \ell_i \leftarrow oc_i, \ell_i\ 1$.

Figure 4.5: *Algorithm 1: priority-aware corridor and layer allocation performed by the DTC at every update.*

the lower-priority drone of the pair (ties broken by identifier) climbs or descends one layer if that layer passes the same gate with a 6 s horizon. Non-emergency drones never enter the emergency layer. A 6 s cooldown prevents oscillation.

Link failure and graceful degradation. The DTC is reached through a link with a configurable outage window. During an outage every request is dropped and counted, the drones continue with their last cached directive, and the age of that directive (staleness) is tracked. When the link returns, directives resume. In map mode the operator can choose a lost-link policy for the fleet: *continue* with the cached directive, *hold* (loiter at the current position until the link returns), *return to base* or *land*. Operator commands sent during an outage are queued for up to 30 s and delivered when the link returns.

4.2.6 Layer Agreement: V2V Negotiation with DTC Pricing

The first map-mode version placed almost all traffic on layers 0 and 1: with 300 drones, 84% of airborne time was on L0/L1 and 1% on L2–L5, and drones queued behind each other under the car-following law while upper layers stayed empty. The layer-agreement protocol spreads traffic over all layers by local negotiation, with the DTC acting as a price setter rather than a dispatcher.

Local cost. The free decision is the layer, within the direction-allowed set $\mathcal{A}_d$. Drone i evaluates, for every candidate layer ℓ,

$$J_i\ell = w_f\,\phi_i\ell\ \ w_p\,\lambda_i\ell\ \ w_e\,c_i\ell\ \ w_s\,\mathbb{1}\ell \neq \ell_i\ \ w_{\text{dir}}\,\mathbb{1}\ell \notin \mathcal{A}_d, \tag{4.12}$$

where $\phi_i\ell = 1 - v^{\text{eff}}v_c$ with $v^{\text{eff}} = \min\big(v_c, g_i\ell - d_0h\big)$ is the speed lost behind the nearest drone ahead on layer ℓ (1 for head-on traffic), $\lambda_i\ell$ is the agreed congestion price, $c_i\ell = \ell - \ell_i\ \frac{H}{H_{\text{ref}}}1\ \ k_b1 - \text{SoC}_i$ is a climb cost that is higher on a low battery, and the last two terms add hysteresis and penalise a wrong-parity layer. The weights are $w_f = 1$, $w_p = 0.6$, $w_e = 0.15$, $w_s = 0.25$, $w_{\text{dir}} = 0.5$ and $k_b = 1$.

Price consensus (dual ascent). Once per second each drone updates its price vector

from its sector neighbours' prices and the load $n_i\ell$ it hears on each layer:

$$\lambda_i\ell \leftarrow \mathrm{clip}_{0,\lambda_{\max}}\Big(M\{\lambda_j\ell : j \in \mathcal{N}_i \cup \{i\}\} \;\alpha\,\frac{n_i\ell - \kappa}{\kappa}\Big), \tag{4.13}$$

with $\alpha = 0.3$, target load $\kappa = 2$ and $\lambda_{\max} = 2$. M is the mean, or the W-MSR trimmed mean with $f = 1$ when the report defence is on, so that one outlying neighbour cannot move the price. This is dual ascent on the slot-load constraint with the dual variable agreed by consensus. When the DTC link is up, the DTC seeds every drone in sector s with $\lambda^{\mathrm{DTC}}_{s,\ell} = n_{s,\ell}C - \theta 1 - \theta$, $\theta = 0.6$, from its global count; when the link is down the drones keep agreeing among themselves.

Local mutual exclusion. Drone i computes $\ell_i^\star = \arg\min_{\ell\in\mathcal{A}_d\cup\{\ell_i\}} J_i\ell$ and its gain $g_i = J_i\ell_i - J_i\ell_i^\star$. If $g_i > 0.15$ it announces $\ell_i^\star, g_i$, and it moves only if no neighbour announcing the same layer has a larger key (gain, then priority, then lower identifier), and only if the layer-transition gate passes. Hence at most one drone enters a given layer per neighbourhood per round. An 8 s cooldown prevents oscillation. Departures use the same cost to pick the cheapest layer with a merge gap instead of waiting for L0/L1.

Stuck drones and DTC settlement. A drone that has stayed below 60% of cruise speed for 4 s (and is not approaching, holding or speed-limited) and cannot move by itself asks the DTC. At its next update, the DTC settles all requests over its global picture by greedily minimising $_i\, w_i K_i \ell_i$ subject to $n_{s,\ell} < C$, the direction rule and the transition gate, with

$$w_i = \left(1 \;\frac{w_{\mathrm{pr}}}{1\;P_i}\right)\left(1 \;w_{\mathrm{bat}} 1 - \mathrm{SoC}_i\right), \tag{4.14}$$

$$K_i\ell = w_d\,\frac{T_i\ell - T_i^0}{T_i^0} \;w_b\,\frac{c_i\ell}{\max\mathrm{SoC}_i, 0.05} \;w_l\,\frac{n_{s,\ell}}{C} \;w_{\mathrm{sw}}\,|\ell - \ell_i|, \tag{4.15}$$

where $T_i\ell$ is the expected remaining flight time on layer ℓ and T_i^0 the free-flow time. Requests are processed in decreasing w_i (highest priority, then lowest battery, chooses first), the standard heuristic for weighted assignment with capacities. If no allowed layer is free, safe and faster, the request is denied.

4.2.7 Resilience: Attack Model and Defence

Threat model. From the attack start time (60 s by default) one or more malicious agents broadcast a falsified position $\mathbf{p}_m = \mathbf{p}_m\ \delta\,\mathbf{e}_\theta$, where δ is a constant offset (5–50 m) or a ramp growing at 1 m/s up to the offset, and $\mathbf{e}_\theta$ is a unit vector whose direction is fixed or derived from the seed. The true state of the malicious agent is never changed, and the same falsified message is sent to every receiver (common delivery). In map mode, custom spoofers can be placed with their own offset and start time, and GNSS quality degrades within 150 m of an active spoofer. Safety is always judged on true positions, so that an attacker cannot hide a collision from the monitor.

Trusted-range check. Each receiver compares the distance implied by a neighbour's report with the range it senses physically (for example by radio ranging). A report whose

implied range differs from the sensed range by more than 3 m is flagged. The flagged neighbour is *not* discarded, because that would make the receiver blind to a real aircraft. Instead its reported position is moved along the reported bearing to the sensed range, so that separation still reacts to it. In the simulation the sensed range is computed from true positions, i.e. an ideal range sensor.

W-MSR filtering. Each receiver also computes a W-MSR estimate with $f = 1$ from its neighbours' reports, coordinate by coordinate. In map mode the corrections from all receivers of a spoofer are fused by W-MSR into an estimate of the spoofer's true position, which is drawn on the map next to its broadcast position, and the same trimmed mean protects the layer prices of Equation (4.13). The operator can finally *quarantine* a spoofer: its reports are ignored network-wide and it is ordered to land.

Undefended baseline. The plain-averaging baseline uses raw reports in the Boids terms with no range check and no trimming. In Weeks 2 and 3 it was shown that a single false report shifts the consensus value of plain averaging, while W-MSR stays within the range of honest values when the robustness condition holds.

4.2.8 Verification: Online STL Monitoring

The core specification is φ_{safe} of Equation (4.1). Its instantaneous robustness is

$$\rho t = \min_{i \neq j} \left(d_{ij} t - d_{\min} \right), \tag{4.16}$$

and over a sliding window of $W = 10$ samples the monitor reports $\min_{t \in \text{window}} \rho t$ (in the corridor benchmark; the map-constrained engine reports ρt at every tick). A value $\rho > 0$ is the spare distance before a violation, $\rho = 0$ is on the boundary (not a violation), and $\rho < 0$ means that the closest pair is $|\rho|$ metres too close. During start-up (fewer than two samples), with fewer than two agents, or with invalid input the monitor returns a neutral value with a status flag rather than a false violation. Derived metrics are the number of violating ticks, the number of violation episodes and the recovery time (time until $\rho > 0$ and stays positive for at least 1 s). An optional liveness specification $\varphi_{\text{live}} = \mathbf{F} \| \mathbf{p}_i - \mathbf{d}_i \| \leq r$ is implemented but not monitored live.

Two distance definitions are used, and this matters for interpreting the results. The corridor benchmark computes d_{ij} as the *horizontal* distance between all pairs, including pairs on different layers. The map-constrained engine computes the *layer-aware* metric distance $d_{ij} = \| \mathbf{p}_i, \ell_i H - \mathbf{p}_j, \ell_j H \|$, counts a loss of separation when two drones on the same layer are closer than $d_{\min}$ (a near miss), and counts a crash when they are closer than 2 m.

4.2.9 Drone Health Model

Every airborne or holding drone carries a deterministic, seeded health model updated every tick. Its parameters per type are listed in Appendix A.

Power. Air density follows from the site elevation (542 m for Hyderabad) and the

ambient temperature. At 35 °C it is 1.074 kg/m^3, compared with 1.148 kg/m^3 at 15 °C, so hot air makes hovering more expensive. With thrust $T = m_{\text{empty}}\ m_{\text{payload}} g$, figure of merit FM, drag area $C_D A_f$, propulsive efficiency η, motor health h_m and avionics power P_{av},

$$P_{\text{mech}} = \frac{T\, v_i v}{\text{FM}}\ \tfrac{1}{2}\rho_a C_D A_f\, v^3\ P_{\text{climb}}, \qquad P_{\text{elec}} = \frac{P_{\text{mech}}}{\eta\, h_m}\ P_{\text{av}}, \tag{4.17}$$

where v is the airspeed (ground velocity minus wind) and each layer climbed (15 m) adds $mg\, v_{\text{climb}}$ for 5 s ($v_{\text{climb}} = 3$ m/s).

Battery. The pack is a 1-RC Thevenin circuit (series resistance R and one R_1C_1 branch with $R_1 = 0.5\,R$ and $R_1C_1 = 20$ s, whose voltage is V_{RC}) with a piecewise-linear lithium-ion OCV curve (3.00 V per cell at 0% to 4.20 V at 100%), resistance increased by ageing and cold, $R = R_{\text{pack}}1\ 0.51 - \text{SoH}1\ \alpha_R 25 - T_{\text{pack}}$, and effective capacity $Q_{\text{eff}} = Q_{\text{nom}}\,\text{SoH}$. The current follows from the quadratic of Chapter 3 with $\text{OCV} - V_{\text{RC}}$ in place of OCV, and $\text{SoC} \leftarrow \text{SoC} - I\,\Delta t 3600\,Q_{\text{eff}}$. If $\text{OCV} - V_{\text{RC}}{}^2 < 4RP_{\text{elec}}$ the pack cannot deliver the power, which is a fault. Remaining usable energy is integrated from the OCV curve down to 5% SoC, and time and range left follow from the current power.

BMS. Each of the n_s series cells has a seeded SoC offset (σ = 0.4%) and resistance factor (σ = 3%), so cell voltages differ under load. The BMS reports a status NORMAL, WARNING, CRITICAL or FAULT, which is the worst of the following codes: low charge (warning below 30%, critical below 15%, fault when any cell falls below 3.0 V), pack over-temperature (55/65/75 °C), motor over-temperature (90/110 °C), cell imbalance (50/100 mV), over-current (80% and 100% of the C-rate limit) and sensor faults. A status only drops after the condition clears by a hysteresis margin (2% SoC, 3 °C, 10 mV), so it does not flicker.

Thermal, link and navigation. Pack and motor temperatures follow first-order lumped energy balances,

$$C_{\text{pack}} T_{\text{pack}} = I^2 R\ I\, V_{\text{RC}}\ q_{\text{fault}} - hAv\, T_{\text{pack}} - T_{\text{amb}}, \qquad hAv = hA_0\ hA_1\sqrt{v}, \tag{4.18}$$

so that a hovering drone cools worse than a cruising one. The DTC link quality follows a log-distance path-loss model with seeded smooth shadowing (σ = 4 dB), and packet loss is a logistic function of RSSI. GNSS horizontal dilution of precision grows near an active spoofer.

Faults. Five faults can be scheduled: capacity loss (remaining charge ×0.7), cell failure (one cell loses 15% SoC and its resistance triples), overheating (extra heat of 60% of hover power into the pack), motor degradation (motor health 0.8) and GNSS degradation.

Onboard responses. Every second each drone checks its energy reserve against the energy needed to finish its route with a 20% margin plus 30 s for landing,

$$E_{\text{need}} = P_{\text{elec}} v_c \frac{L_{\text{rem}}}{v_c} 1\ 0.2\ P_{\text{elec}} 0\, t_{\text{land}}. \tag{4.19}$$

If the remaining energy is below E_{need}, the drone diverts to the nearest pad along

the corridor graph that it can still reach, and lands in place if none is reachable. A CRITICAL drone lands at the nearest reachable pad; a FAULT drone lands in place immediately; an overheating drone is limited to 60% of cruise speed and lands if it stays CRITICAL for 30 s. A pre-flight check keeps a drone on its pad if it is already CRITICAL or cannot reach its destination.

DTC rule for unhealthy drones. A CRITICAL or FAULT drone is treated like an emergency for the right of way to land: its priority becomes 0 in DTC allocation, in ADMM (weight 10) and in tactical shifts, so that other drones yield; it is *not* moved to the emergency layer, because a climb costs energy; and its landing pad is reserved, with departures from that pad held until it has landed. Vertiport arrival sequencing and take-off and landing scheduling have been studied in more depth (Kleinbekman et al., 2018; KrisshnaKumar et al., 2023); MASooda uses only this simple reservation rule. Between two unhealthy drones, the one with lower SoC goes first.

4.2.10 Operator Console and Scenario Engine

The DTC console issues air-traffic-control style commands to selected drones: hold and resume, ground now, return to base, layer change, reroute through a via point, drone or sector speed limits, priority changes, quarantine, battery swap or charge, lost-link policy, temporary flight restrictions (TFRs) drawn on the map, sector ground stops and fleet-level health actions. Every movement command is first checked against the drone's remaining energy, and an infeasible command is rejected with the nearest reachable pad. Commands travel over the DTC link and are acknowledged, queued, failed or expired. A scenario file contains a header (map, seed, fleet and mix, duration, architecture), conditions (weather, attack, outage), custom drones and a timed command script. Scripted commands are applied at their tick exactly like operator commands, so every preset replays identically. Long-duration operations add repeated sorties with turnaround times and battery-service pads with finite capacity and first-in-first-out queues.

4.2.11 Architectures Compared

All architectures share the dynamics, the world and the monitor, and start from the same initial conditions, verified by a SHA-256 hash of every agent's identifier, type, priority, rounded position, destination, layer and corridor.

- **Hierarchical (MASooda):** DTC allocation, local Boids and car-following, ADMM, layer agreement and the report defence.
- **Centralised:** a single controller sees every reported state and computes every velocity directly: centreline tracking, a time-headway slow-down behind the leader on the same layer, and a priority-split pairwise separation push (the lower-priority agent takes all of it, equal priorities share it). It runs no Boids, no ADMM and no defence, and trusts every report. During an outage agents keep their last command.
- **Decentralised:** no DTC; local behaviour, ADMM and the defence (map mode only).

- **Decentralised undefended:** no DTC, local behaviour and ADMM on raw, possibly falsified reports; agents keep their initial corridor.

4.2.12 Theoretical Analysis

Proposition 4.1 (Conservative linearisation)**.** *If next positions* $\mathbf{x}_i, \mathbf{x}_j$ *satisfy the linearised constraint* (4.7)*, then* $\|\mathbf{x}_i - \mathbf{x}_j\| \geq d_{safe} > d_{\min}$.

Proof. $\mathbf{n}_{ij}$ is a unit vector, so by the Cauchy–Schwarz inequality $\|\mathbf{x}_i - \mathbf{x}_j\| \geq \langle \mathbf{n}_{ij}, \mathbf{x}_i - \mathbf{x}_j \rangle \geq d_{\text{safe}}$. □

Proposition 4.2 (Convexity and convergence of the ADMM subproblem)**.** *For a fixed conflict group, problem* (4.6)–(4.8) *has a strongly convex objective (all* $w_i > 0$*) and convex constraints (half-spaces and balls). The consensus splitting has closed, proper, convex terms, so whenever the problem is feasible the ADMM iterates converge to its unique optimum and the residuals converge to zero (Boyd et al., 2011).*

Remark 4.1 (What is and is not guaranteed). The two propositions guarantee separation at the *predicted next positions of the agents in the solved group*, and only when ADMM converges and the problem is feasible. They do not guarantee separation of the executed trajectory. The prediction assumes that the resolved velocity is applied in one step, whereas the dynamics track the commanded velocity with a time constant and acceleration limit; pairs outside the detection radius, and pairs between different groups, are not constrained; and the problem can be infeasible when the speed balls are too small to open the gap in one tick. ADMM is therefore reported as conflict *mitigation*, not as a safety guarantee. The STL monitor measures the outcome independently.

Remark 4.2 (Priority under ADMM). For a single active pair constraint with the speed limits inactive, the KKT conditions of (4.6) give displacements $w_i\mathbf{x}_i - \mathbf{p}_i = \mu\,\mathbf{n}_{ij}$ and $w_j\mathbf{x}_j - \mathbf{p}_j = -\mu\,\mathbf{n}_{ij}$ with the same multiplier $\mu \geq 0$. The deviations are therefore inversely proportional to the weights: against a delivery drone ($w = 1$), an emergency drone ($w = 10$) absorbs only 111 of the manoeuvre.

Remark 4.3 (Resilience). W-MSR guarantees resilient consensus only when the communication graph is sufficiently robust, $2f$ 1-robust for an f-local adversary (LeBlanc et al., 2013). The radius graph of a moving fleet does not always meet this condition; in sparse areas a drone may have fewer than $2f$ 1 neighbours, in which case the filter falls back to the median of its own and the received values. The integrated results are therefore empirical. The trusted-range check provides a complementary, physics-based signal that does not depend on graph robustness.

Remark 4.4 (Monitor soundness). Because the robustness (4.16) is computed on true positions at every tick, $\rho t_k < 0$ if and only if some pair violates $d_{\min}$ at tick t_k. The monitor is sound and complete at the sampling instants, but it is not a continuous-time guarantee between samples. At 15 m/s and $\Delta t = 0.1$ s, two drones can close by at most 3 m between samples.

Remark 4.5 (Safety gate). In map mode, a DTC directive or tactical shift moves a drone into layer ℓ only if no drone on ℓ is within $3d_{\min}$ and no CPA on ℓ below $2d_{\min}$ is predicted within the horizon. Under constant velocities over the horizon, a layer change therefore never creates an immediate loss of separation on the target layer.

4.2.13 Complexity Analysis

Table 4.1 summarises the cost per tick of each component for N airborne agents with average degree d. With the uniform-grid spatial index (cell size equal to the communication radius), neighbour queries are local and the graph, the Boids terms, the CPA checks and the separation monitor all become close to linear in N for bounded density. ADMM is solved per conflict group: one iteration costs time linear in the number of agents n_g and pair constraints e_g of the group, and the number of iterations is bounded by the iteration limit. The DTC runs only every 20 ticks and sorts agents once. The spatial index changes no result; it only makes large fleets faster, which is verified by a regression test.

Component	**Naive cost**	**Cost with grid index**
V2V graph (radius)	ON^2	ONd
Local behaviour (Boids, CPA, car-following)	ONd	ONd
Conflict detection and grouping	ON^2	ONd
ADMM, per group and iteration	$On_g\ e_g$	same; ≤ 50 iterations
DTC allocation (every 2 s)	$ON \log N\ \ NL$	same
W-MSR filter, per receiver	$Od \log d$	same
Layer agreement (every 1 s)	$ONLd$	same
STL robustness (min separation)	ON^2	ONd same-layer pairs
Health model and BMS	ONn_s	same

Table 4.1: *Per-tick computational cost of the components. n_g, e_g: agents and pair constraints in a conflict group; L: layers; n_s: series cells per pack.*

In practice, the measured ADMM cost in the corridor benchmark is about 1–2 ms per tick for 30–75 agents (Section 5.1.2), and the map-constrained engine runs a 500-drone fleet at 3.4 times real time on a 2-vCPU 2.1 GHz Xeon server, which set the upper fleet limit of the public demonstration.

5

Simulations & Demonstrations

This chapter describes how the system was implemented and tested, and reports three groups of results: the integration milestones of Weeks 3–5, the six-experiment campaign on the corridor benchmark (Week 6), and the map-constrained live airspace with its scripted scenarios. Because the corridor campaign and the live airspace measure separation differently (Section 5.1.2), their safety numbers must not be compared directly.

5.1 Simulation Results

5.1.1 Simulation Setup

Implementation. The simulator is written in Python 3.10+ with NumPy for numerics, PyYAML for configuration and Matplotlib for analysis figures. The live server uses the Python standard-library HTTP server, with no web framework, and streams compact frames as JSON. The browser front end is written in plain HTML, CSS and JavaScript on a 2D canvas, without frameworks. The live demonstration is deployed publicly on Render. All model parameters, limits and experiment matrices live in one configuration file, and no module hard-codes a simulation parameter. Visualisation-only parameters (colours, figure sizes, refresh rates) are kept in a separate section that the simulation never reads, so changing them cannot change a result.

Base parameters. Table 5.1 lists the base configuration shared by all experiments. The full parameter set, including the health model, is given in Appendix A.

Experiment campaign. The Week 6 campaign consists of six experiments (Table 5.2), 141 runs in total, generated from declarative experiment matrices by one runner. Each run stores a standard row of 22 columns (14 metrics), per-tick telemetry, fleet snapshots every second and a manifest with the commit, configuration hash and seeds. Summary tables report the mean over seeds $\pm$ the 95% t-interval half-width. A result group is flagged as *weak* if it has fewer than three finite values or a coefficient of variation above 0.5. Fourteen groups were flagged; they are listed but were not re-run.

Reproducibility. Every architecture starts from bitwise-identical initial conditions,

Parameter	Value	Role
Number of agents	60 (30–75 in sweeps)	fleet size
Type mix	10% / 30% / 60%	emergency / eVTOL / delivery
Time step, duration	0.1 s, 180 s	simulation clock
Communication	radius 40 m (or $k = 6$)	V2V graph
Minimum separation $d_{\min}$	8 m	safety specification
Speed, acceleration limits	15 m/s, 4 m/s^2	dynamics
Cruise speed v_c	12 m/s	local behaviour
Corridors	3 × 20 m wide, 50 m apart, 1500 m long	benchmark world
Corridor capacity	15 agents per layer	DTC allocation
ADMM tolerance, iteration limit, ρ	10^{-3}, 50, 1.0	conflict resolution
ADMM detection radius	20 m	conflict detection
DTC update period	2.0 s	global tier
Altitude layers	3 (benchmark, 30 m apart), 7 (live, 15 m apart, L0 at 30 m)	2.5D airspace
STL period, window	0.1 s, 10 samples	monitor
Attack start, offset	60 s, 25 m (5–50 m sweep)	threat model
W-MSR parameter f, range tolerance	1, 3 m	defence
DTC outage window	90–120 s (when enabled)	link failure
Seeds	42, 43, 44 (+45, 46 for E4)	repetitions

Table 5.1: *Base simulation configuration.*

checked by hashing the initial fleet. A single script regenerates every experiment, figure and table, with a quick mode (one seed, 12 agents, 20 s) for smoke tests. Two consecutive quick runs from a clean copy of the repository produced zero differences in the run tables, and all 58 expected artefacts were produced.

Testing. Testing followed four levels: unit tests of single modules, integration tests through the defined interfaces, system tests of complete scenarios, and experiments. The repository contains 421 test functions. They cover, among others, dynamics clamping, two-agent and N-agent ADMM on head-on, crossing and overtaking scenarios, conflict detection equivalence across seeds, W-MSR trimming, the STL monitor on hand-crafted safe and unsafe trajectories, the overseer link outage, the tube and geofence invariants of the route engine (every airborne drone stays inside its tube and the geofence on every tick), the equivalence of the spatial index, the health model, the operator console, layer agreement and every scenario preset. Table 5.3 shows the failure-mode test matrix defined in the workflow. Its expected recoveries are those planned in the workflow; in the implementation, falsified broadcasts (case 4) are handled mainly by the trusted-range check, and a congested corridor (case 7) is counted as overflow rather than rerouted or throttled.

Hardware. The campaign was run on a laptop with an Intel Core i5-12500H; the recorded wall time of the final invocation of every experiment folder was 7.94 minutes.

Exp.	Question	Conditions	Runs
E1	ADMM scaling	hierarchical; $N \in \{30, 45, 60, 75\}$; iteration limit $\{5, 10, 20, 50, 100\}$ at $N = 60$	24
E2	merging and throughput	hierarchical; $N \in \{30, 45, 60, 75\}$	12
E3	priority compliance	hierarchical; one emergency agent vs a paired control run	24
E4	resilience under attack	three architectures; attack off/on; offsets 5–50 m; $N = 60$	57
E5	STL robustness	all architectures and densities (pooled with E1–E4)	18
E6	graceful degradation	hierarchical; DTC link up vs down 90–120 s; $N = 60$	6

Table 5.2: *Week 6 experiment matrix. Run counts are those stored for each experiment folder; E5 adds only the combinations that E1–E4 lack.*

5.1.2 Results & Discussions

Integration Milestones (Weeks 3–5)

The Week 3 checkpoint ran a 30-agent, three-corridor simulation for 60 simulated seconds with local behaviour, corridor allocation, the priority-aware DTC, N-agent ADMM, trajectory logging and live STL monitoring, without crashes or invalid states. The Week 4 end-to-end MVP ran the full pipeline with 60 agents for 180 s, with one malicious agent from 60 s and a DTC outage from 90 to 120 s. It produced 2001 ADMM events, a minimum separation of 5.44 m, 377 violating ticks in 12 episodes (all recovered, mean recovery 3.3 s), 56 of 60 arrivals, a maximum directive staleness of 31.9 s and no component failures. The defence corrected 3981 falsified reports.

Three Week 4–5 results shaped the later work:

- **ADMM reduces, but does not remove, merge conflicts.** In a 60-agent, 120 s merge test, ADMM reduced the ticks with a pair closer than 8 m from 388 to 255 (-34%), but the minimum distance was 0.41 m with ADMM and 1.31 m without. This motivated the fallback rule, the intact conflict components and the honest framing of ADMM as mitigation.
- **The defence is precise.** In the Week 4 attack demonstration (60 agents, offsets of 10, 25 and 50 m), the range check flagged between 3010 and 3417 falsified reports per run and wrongly flagged no honest report.
- **The centralised baseline is safe but trusting.** Across three seeds it delivered all 60 agents, with a minimum separation of 5.2–6.6 m and every violation episode recovered in 1.7–4.2 s. On seed 44 the attack raised its violation ticks from 70 to 109 and lowered its minimum separation from 6.58 to 5.49 m, showing its exposure to unverified reports.

#	Failure case	Observable metric	Expected recovery
1	normal communication	graph connectivity ratio	baseline operation
2	missing neighbour information	validity of local output	fall back to goal seeking
3	DTC link failure	directive staleness	continue with cached directive
4	falsified position broadcast	malicious reports filtered	W-MSR limits the influence
5	priority conflict	priority-compliance rate	priority ordering kept
6	ADMM timeout	non-converged events	fall back to Boids separation
7	congested corridor	occupancy vs capacity	rerouting or entry throttling
8	agents entering/leaving range	graph churn	neighbour sets update, no crash
9	STL safety violation	robustness value	flagged and logged
10	component failure	fallback count	tick continues with fallback

Table 5.3: *Failure-mode test matrix.*

A Note on the Safety Metric of the Corridor Benchmark

All Week 6 corridor results are marked *provisional* by the team, and one property of the metric dominates their safety numbers. The corridor benchmark computes separation in the horizontal plane over *all* pairs, including pairs on different altitude layers, whereas conflict detection, ADMM and the DTC treat agents on different layers as separated. The hierarchical architecture is the only one in the benchmark that uses layers (Figure 5.1): its DTC places emergencies on the reserved layer and spreads other agents over layers 0 and 1 from the start. Drones on different layers can therefore be directly above each other, which the horizontal metric records as a separation of almost 0 m and a robustness of almost −8 m. The robustness time series confirm this: the hierarchical curve falls below zero within the first three seconds, when agents are still in the entry zone and long before any merge or attack (Figure 5.7). The team's audit identifies a layer-aware separation metric as the fix; the map-constrained engine already uses it. The hierarchical violation counts below should be read with this in mind; the centralised and decentralised architectures keep all agents on one layer, so their numbers are not affected.

E1: ADMM Scaling

Table 5.4 and Figures 5.2–5.3 show that ADMM effort per solve is independent of fleet size. The mean number of iterations stays at about 10 (95th percentile 14–15) for every N from 30 to 75, every solve converged within the default limit of 50, and the mean solve time stays at about 1 ms. The number of solver calls grows roughly linearly with N (947 to 2126 per run), and so does the total ADMM time per tick (0.77 ms to 1.79 ms), which

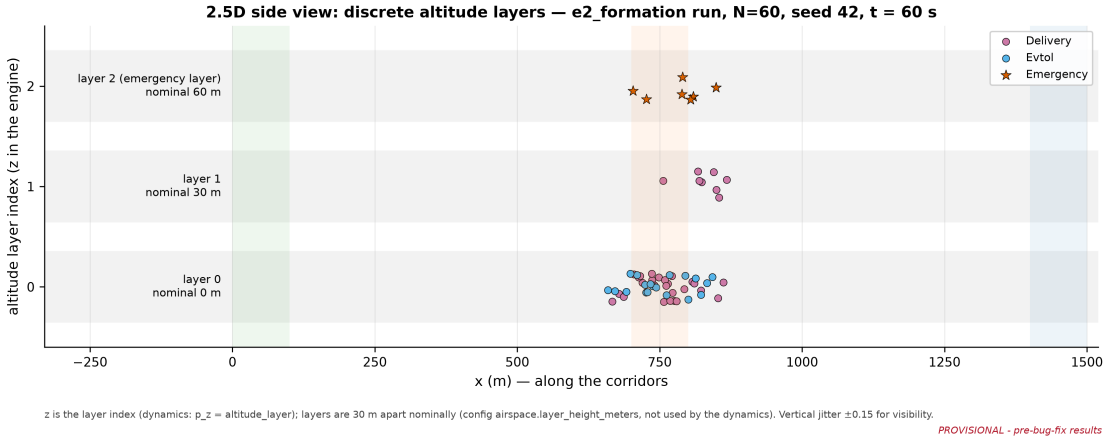


Figure 5.1: *Side view of the corridor benchmark (hierarchical, 60 agents, $t = 60$ s). Emergencies occupy the reserved layer 2, and delivery agents are spread over layers 0 and 1. Pairs stacked on different layers have near-zero horizontal distance.*

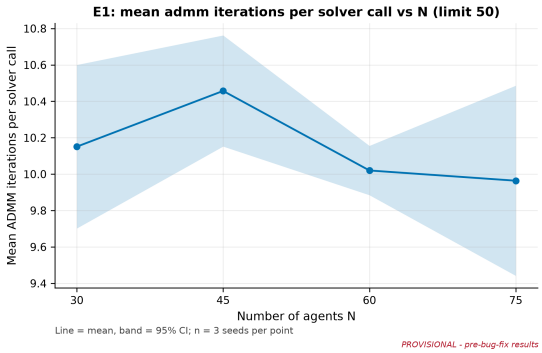
is far below the 100 ms tick. The largest solved group had five agents, which was the group-size cap in force when the campaign ran; the cap was removed afterwards so that every conflict component is solved intact (Section 4.2.4), and group sizes and timings must be re-measured with that change. The iteration-limit sweep at $N = 60$ shows that a limit of 5 gives only 7% convergence and a limit of 10 gives 78%, while 20 iterations already give 99.7%; limits above 20 change nothing. The default of 50 therefore has a comfortable margin. Changing the limit did not change the number of violating ticks, which is consistent with the metric note above.

N	solver calls	mean iter.	p95 iter.	converged	solve (ms)	ADMM/tick (ms)
30	947 ± 300	10.15 ± 0.45	14.3	1.00	0.90 ± 0.17	0.77 ± 0.09
45	1360 ± 530	10.46 ± 0.31	15.0	1.00	1.06 ± 0.09	1.21 ± 0.30
60	1809 ± 640	10.02 ± 0.14	15.0	1.00	1.03 ± 0.12	1.55 ± 0.29
75	2126 ± 330	9.96 ± 0.52	15.0	1.00	1.00 ± 0.14	1.79 ± 0.13
Iteration limit at $N = 60$ (converged fraction): 5: 0.07, 10: 0.78, 20: 0.997, 50: 1.00, 100: 1.00						

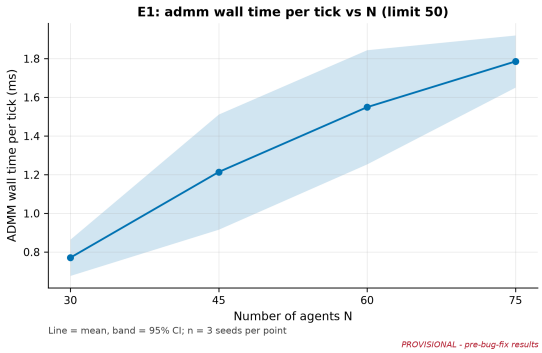
Table 5.4: *E1: ADMM scaling (hierarchical, no attack, iteration limit 50; mean ± 95% CI over three seeds).*

E2: Corridor Merging, Throughput and Utilisation

Throughput grows linearly with demand, from 0.156 arrivals/s at $N = 30$ to 0.417 arrivals/s at $N = 75$, where all 75 agents arrive within 180 s (Table 5.5). Mean corridor utilisation grows from about 0.5–0.65 of capacity at $N = 30$ to 1.3–1.6 at $N = 75$. The centre corridor is always the most loaded because it receives merging streams from both sides, and with 60 or more agents the corridors hold more than 15 agents for thousands of ticks. This is not DTC overflow: the DTC's capacity of 15 applies per (corridor, layer), and each corridor had two normal layers. The merge success rate is close to zero at every density. The team attributes this to the definition of a successful merge rather than to



(a) *Iterations per solve versus fleet size.*



(b) *ADMM time per tick versus fleet size.*

Figure 5.2: *E1: ADMM effort per solve is flat in N; total time per tick grows roughly linearly with the number of conflicts.*

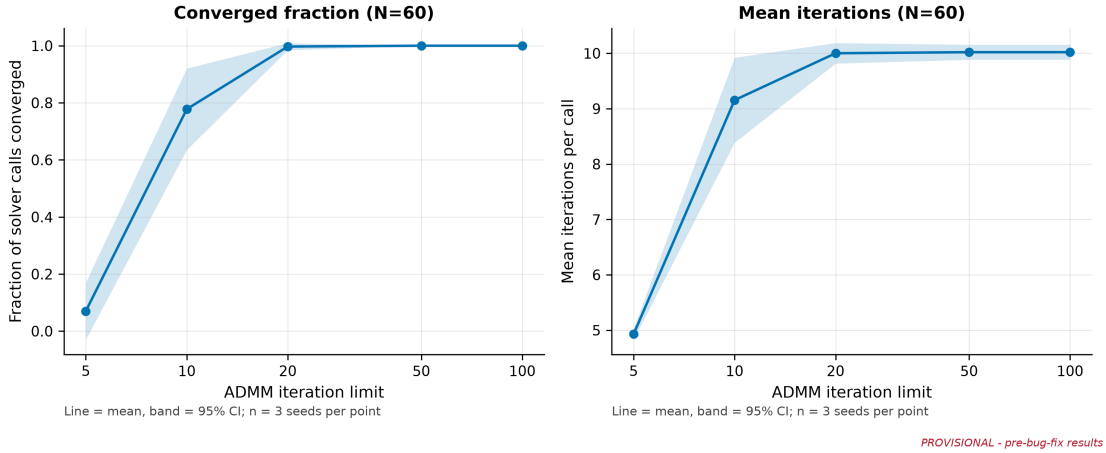
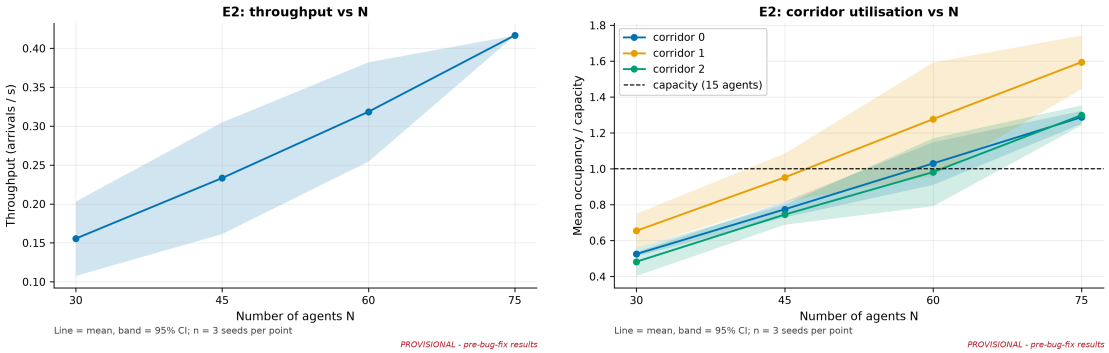


Figure 5.3: *E1: converged fraction and mean iterations versus the ADMM iteration limit at $N = 60$. Convergence saturates at about 20 iterations.*

merging behaviour: an agent counts as a success only if it is already inside its target corridor at the downstream edge of the merge zone. A simple calculation supports this. Crossing to a neighbouring corridor needs about 40 m of lateral motion, which takes 20 s at the 2 m/s lateral speed limit, while an agent at cruise speed spends only about 8 s in the 100 m merge zone. The merge definition is listed as an open issue.

N	merge attempts	throughput (1/s)	arrivals	util. C0	util. C1	util. C2
30	21.0	0.156 ± 0.048	28.0	0.53	0.65	0.48
45	29.7	0.233 ± 0.072	42.0	0.77	0.95	0.75
60	40.0	0.319 ± 0.064	57.3	1.03	1.28	0.98
75	47.7	0.417 ± 0.000	75.0	1.29	1.60	1.30

Table 5.5: *E2: merging, throughput and mean corridor utilisation (occupancy divided by the capacity of 15), hierarchical architecture.*



(a) *Throughput versus fleet size.* **(b)** *Corridor utilisation versus fleet size.*

Figure 5.4: *E2: throughput and corridor utilisation in the corridor benchmark.*

E3: Priority Compliance

In E3 one emergency agent is injected into traffic, and every run is paired with a control run in which the same agent is an ordinary delivery agent. The emergency agent reaches the exit 8.7 s later than free flow on average and saves 10–36 s compared with its control run (Table 5.6); Figure 5.5 shows a typical track, in which the emergency agent merges directly to its corridor and stays close to the free-flow line while its control twin is slowed in the merge zone. The delay imposed on other traffic is within about ±2 s, with no clear trend. The compliance rate, defined as the fraction of DTC checks at which the emergency agent's assigned corridor holds at most 15 agents, is 1.0 at $N = 30$ but collapses to about zero at 45 agents and above. This reflects the metric definition, not a failure to prioritise: it compares the emergency agent's whole-corridor load (all layers) with the per-layer capacity of 15, so it fails as soon as a corridor holds more than 15 agents. The emergency delay is identical across N because the same agents are selected at every density for each seed.

N	compliance	delay (s)	saved (s)	imposed (s)	travel E / C (s)
30	1.00	8.7	22.5	−0.15	125.2 / 147.7
45	0.01	8.7	10.0	0.59	125.2 / 138.8
60	0.00	8.7	35.6	1.81	125.2 / 160.8
75	0.00	8.7	11.4	−0.53	125.2 / 136.6

Table 5.6: *E3: emergency priority (means over seeds 42–44): compliance rate, emergency delay against free flow, time saved against the control run, delay imposed on other agents, and travel times of the emergency agent (E) and of the same agent in the control run (C). Confidence intervals are wide (Section 5.2).*

E4: Resilience under Attack

E4 compares the three architectures with one agent falsifying its position from $t = 60$ s, over five seeds, plus an offset sweep (Table 5.7, Figure 5.6). Throughput is essentially unchanged by the attack in every architecture. For the centralised architecture, which trusts every report, the attack increases violating ticks from 117 to 141 and deepens the

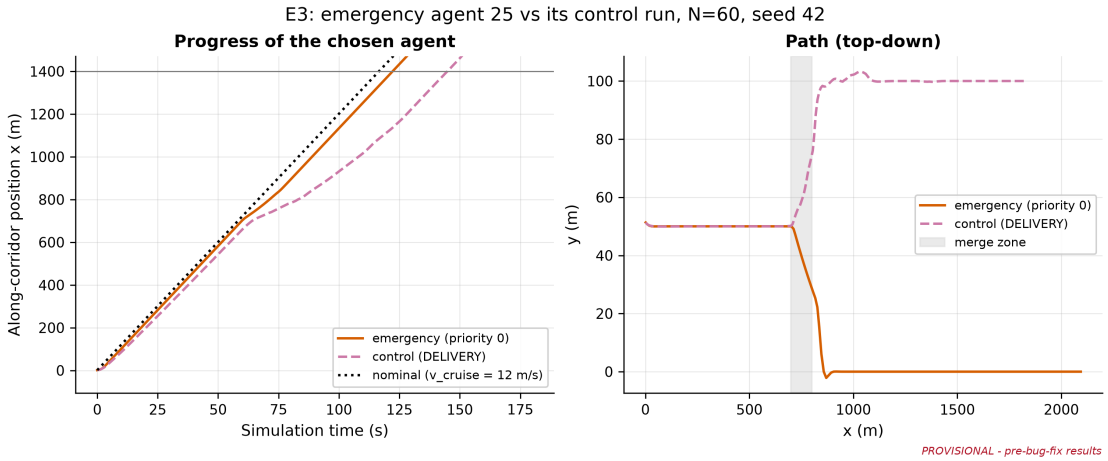


Figure 5.5: *E3: progress and path of an emergency agent and of the same agent as an ordinary delivery drone in the paired control run ($N = 60$, seed 42).*

minimum robustness from −1.83 to −2.48 m at a 25 m offset, with 24 violating ticks after the attack starts. The undefended decentralised architecture has no violations without attack; at 25 m it records 76 violating ticks on average (driven by one seed) and at 50 m 432, all of them after the attack starts. The hierarchical numbers are dominated by the metric artefact and do not change with the attack. A paired analysis found that for all three architectures the 95% interval of the attack-induced difference includes zero for all four metrics, so with one attacker among 60 agents the effect is not statistically significant at five seeds. The trend nevertheless points in the expected direction for the undefended decentralised architecture, whose violating ticks grew from 0 at 5–10 m to 76 at 25 m and 432 at 50 m; the centralised architecture showed no trend with the offset.

Architecture	Attack	throughput (1/s)	violating ticks	min ρ (m)
Centralised	off	0.333	117 ± 64	-1.83 ± 0.78
Centralised	on (25 m)	0.333	141 ± 50	-2.48 ± 0.39
Decentralised undefended	off	0.333	0	0.04 ± 0.04
Decentralised undefended	on (25 m)	0.333	76 ± 210	-0.09 ± 0.37
Decentralised undefended	on (50 m)	0.333	432 ± 1100	-0.65 ± 1.7
Hierarchical*	off	0.321 ± 0.024	1777 ± 6	−8.00
Hierarchical*	on (25 m)	0.310 ± 0.030	1777 ± 6	−8.00

Table 5.7: *E4: architectures under a position-falsification attack ($N = 60$, five seeds; 50 m offset three seeds). *Dominated by the horizontal-metric artefact of Section 5.1.2.*

E5: STL Robustness across Architectures and Densities

Pooling the nominal runs of all experiments (Figure 5.8), the decentralised (undefended) architecture never violates separation without an attack, with a minimum robustness of 0.04–0.14 m and a 5th-percentile robustness of 1.6–3.5 m. The centralised architecture violates in 3% of ticks at $N = 30$ and in 10% at $N = 75$, with a mean recovery of 2.4–4.2 s, which shows its loss of margin as density grows. The hierarchical violating

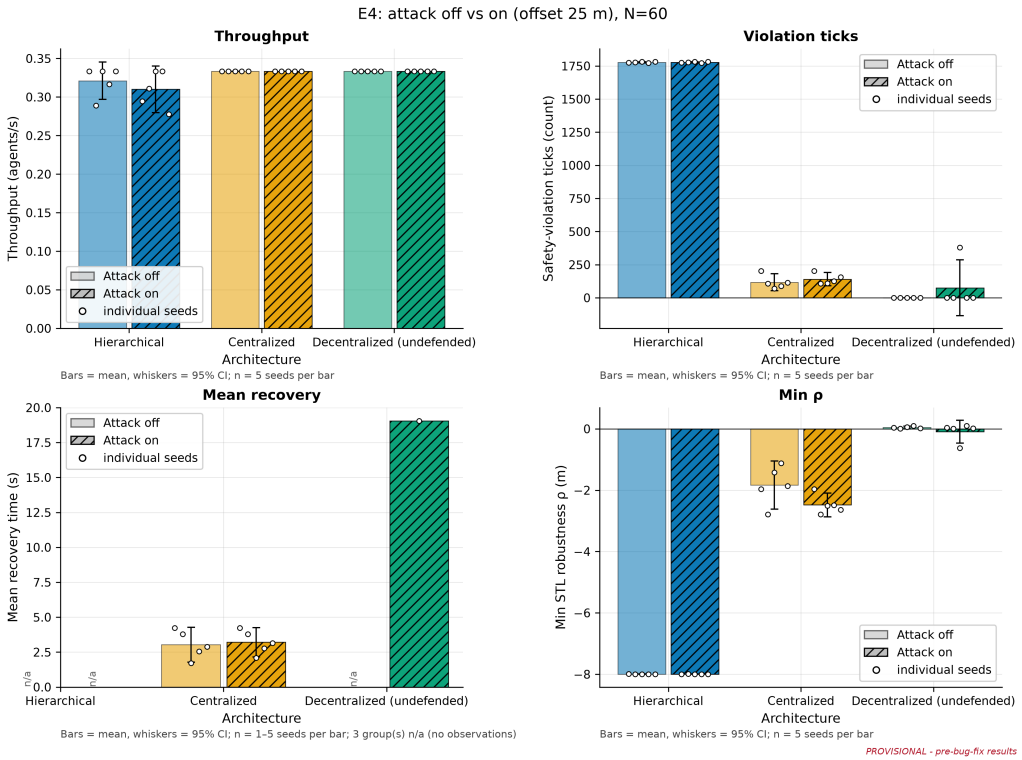


Figure 5.6: *E4: throughput, violating ticks, mean recovery time and minimum robustness for the three architectures, attack off and on (25 m offset). Points are individual seeds.*

fraction (46% at $N = 30$, 99% at $N = 75$) again reflects stacked agents on different layers rather than near-collisions. The STL monitor itself behaved as specified in every run: it flagged every negative sample, reported episodes and recovery times, and never reported a violation during start-up.

E6: Graceful Degradation under DTC Link Failure

With the DTC link down from 90 to 120 s, every one of the 15 directive requests in the window was dropped and the age of the cached directive rose linearly to 31.9 s before resetting when the link returned (Figure 5.9). The system kept running without errors. However, the trajectories of the outage runs and the normal runs were identical for every seed (maximum position difference 0 m), so violations, throughput and robustness are identical and the degradation score is exactly 1. The most likely reason is that the directives in force at 90 s were still valid: by then corridor assignments had settled, so a fresh directive would have been the same as the cached one. E6 therefore shows that the cached-directive mechanism works, but not yet how the system degrades when directives *need* to change during an outage. The map-constrained scenarios address this with lost-link policies and a scripted outage during congestion (Table 5.9).

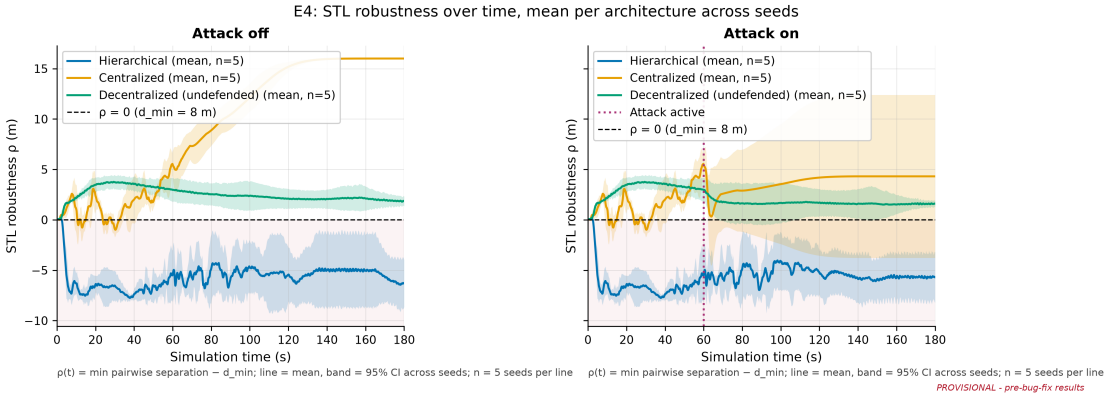


Figure 5.7: *E4: STL robustness over time, mean and 95% band across five seeds. The hierarchical curve drops below zero within about 2.5 s, before any merge or attack, which reveals the cross-layer artefact.*

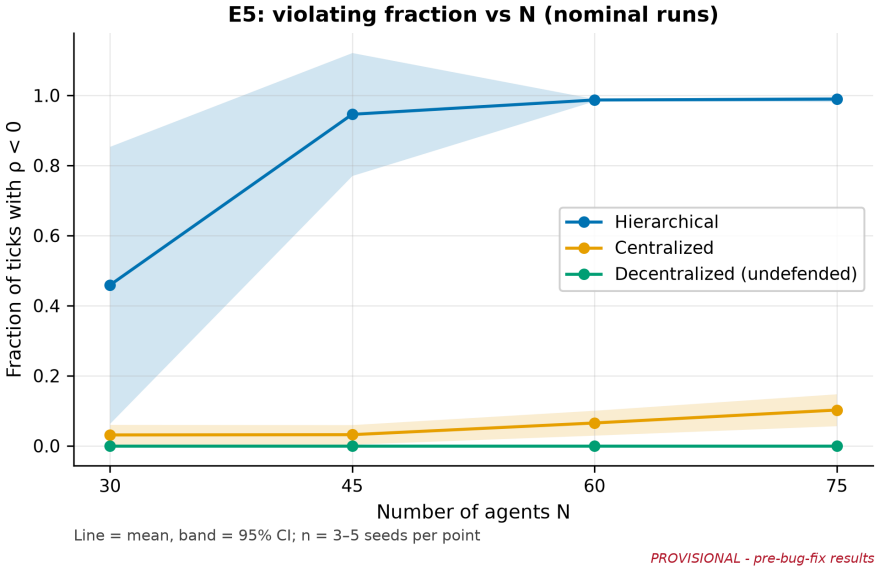


Figure 5.8: *E5: fraction of ticks with $\rho < 0$ per architecture and fleet size (nominal runs).*

Map-Constrained Live Airspace

Layer-aware verification. The route engine was verified with a seed sweep of 30 runs of 180 s: five seeds with 30, 60 and 90 drones, plus attack, outage and undefended variants at 60 drones. With the layer-aware metric, 29 of the 30 runs had **zero** separation violations. The exception was a 90-drone stress run (seed 11) with 5 ticks at a minimum of 5.6 m, caused by a head-on meeting on the shared emergency layer, which the STL monitor flagged correctly. All six quick-select scenarios of the user interface (60 drones, 240 s) also had zero violations.

Layer agreement. Table 5.8 shows the effect of the layer-agreement protocol on the IITH map with 300 drones over 240 s (seed 42, hierarchical). Fixing the DTC allocator alone moved only 13% of airborne time to the upper layers. Adding departure spreading, V2V agreement and DTC pricing raised the share of layers 2–5 from 1% to 46%, increased arrivals from 181 to 246 (+36%), cut the time that non-emergency drones spent below

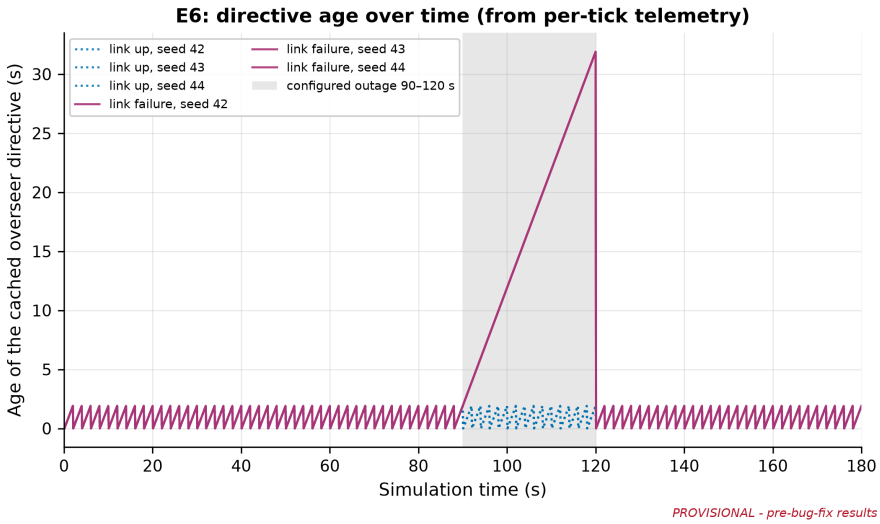


Figure 5.9: *E6: age of the cached DTC directive with the link up (dotted) and with a link failure from 90 to 120 s (solid).*

half of cruise speed from 6.4% to 2.3%, reduced ADMM conflict events from 7549 to 2189 (−71%) and removed the only near miss, raising the minimum separation from 7.6 to 12.8 m. During the run, drones made 52 agreed layer switches, yielded 21 times to a neighbour with a larger gain and spread 142 departures above the lowest layer; of 21 stuck requests, the DTC resolved one and denied 20 for lack of a free, safe and faster layer. The price paid is computation: 75 s instead of 55 s for the 240 s run.

	Before	**DTC allocator fixed**	**+ spreading, V2V agreement, pricing**
Airborne time on L0 / L1 / L2–L5 / L6 (%)	41 / 43 / 1 / 15	33 / 39 / 13 / 14	20 / 23 / 46 / 11
Arrivals in 240 s	181	183	**246**
Non-emergency time below 50% cruise	6.4%	4.8%	**2.3%**
Near misses, min separation	1, 7.6 m	1, 7.6 m	**0, 12.8 m**
ADMM conflict events	7549	4166	**2189**
Tactical layer shifts	58	39	23
Compute time	55 s	65 s	75 s

Table 5.8: *Layer agreement on the IITH campus map (300 drones, 240 s, seed 42, hierarchical).*

Scripted scenarios. Table 5.9 summarises the headless results of the ten scripted scenarios on the IITH map (seed 42); their timelines are given in Appendix B.

Three findings stand out. First, the DTC outage scenario shows a real trade-off between lost-link policies that E6 could not show: continuing with the cached directive delivered the most, holding protected the airspace but cost 77% more energy and delayed emergencies by 51 s, and returning to base delivered least. Second, the heat-wave comparison shows that hot weather alone keeps pack temperatures below the 55 °C warning on short campus flights, while it raises energy per delivery by about 4% and motor temperatures

Scenario	Result
Medical emergency (120 drones)	emergency compliance 100%, mean emergency delay 4.5 s, 0 near misses
Vertiport congestion (200)	223 ground holds, 161 of 200 delivered in 300 s, 0 near misses
VIP restricted area (120)	17 restricted-area events, departures from the closed pads held, 0 near misses
GPS spoofing (100)	5 reports corrected, both spoofers quarantined at 110 s, 1 near miss during the attack
DTC outage (100)	*continue*: 91 delivered; *hold*: 76 delivered, energy +77%, emergency delay 51 s; *return to base*: 36 delivered (the rest went home)
Weather cell (120)	120 restricted-area events, 3 drones caught inside landed, 0 near misses
Non-cooperative intruder (80)	detected by onboard sensing, never seen by the DTC or ADMM, 0 crashes
Battery failure (100)	cell-failure drone CRITICAL → right of way → diverted and landed; capacity-loss drone diverted on low reserve; 0 depleted
Heat wave, 44 vs 30 °C (150)	energy per delivery +4%; maximum motor temperature 69.8 vs 56.2 °C
60-minute operations (60)	1469 deliveries, 137 battery swaps, swap queue ≤ 5, 0 forced landings, 0 depleted

Table 5.9: *Headless results of the scripted scenarios on the IITH campus (seed 42).*

by 14 K; the scheduled overheating faults dominate pack temperature in both variants. Third, the health responses work end to end: in the battery-failure scenario, a failed cell raised the BMS status to CRITICAL, the DTC gave the drone the right of way and reserved its pad, and the drone diverted and landed, while a second drone with a capacity loss diverted on its low-reserve check, with no drone running out of energy.

Health model plausibility. Table 5.10 lists the steady-state figures that the default parameters produce at 35 °C in Hyderabad. The endurances of 21–48 minutes and ranges of 17–41 km are in the range of commercial delivery multirotors and published small-eVTOL estimates (Stolaroff et al., 2018; Yang et al., 2021). At 42 °C, hover power is about 1% higher than at 35 °C and about 4% higher than at 15 °C, because of lower air density.

	Delivery	Emergency	eVTOL
Hover / cruise power	1.38 / 1.06 kW	1.62 / 0.98 kW	86.0 / 60.8 kW
Endurance hover / cruise	21.1 / 27.5 min	28.6 / 47.5 min	16.9 / 23.8 min
Range at cruise	17.8 km	41.1 km	17.2 km
Cruise current / pack heat	47.8 A / 69 W	22.0 A / 29 W	152 A / 1.73 kW
Steady pack / motor temperature rise	+10.9 / +18.3 K	+4.3 / +12.6 K	+11.8 / +19.1 K

Table 5.10: *Steady-state outputs of the drone health model with default parameters at 35 °C and 542 m elevation (SoH 100%, energy counted down to 5% SoC, pack heat including the RC branch).*

Live Demonstration

The final demonstration streams the map-constrained engine to the browser (Figure 5.10). The server starts a run and returns frames while the simulator computes them, about 10–15 times faster than real time for 60 drones; while a page is watching, the simulator stays at most 5 s ahead of playback so that operator commands land near the time on screen. The page offers the IITH and Hyderabad maps; the four architectures; the ten scenario presets with a timeline and side-by-side comparison of variants; a 2.5D view of the layers around the selected drone with its own event feed (ADMM negotiations, DTC directives, safety-gate deferrals, tactical shifts, spoofed reports filtered, landings); a network-theory panel of the DTC hub and V2V graph with degree, density, transitivity, centrality, components and algebraic connectivity over time; an ADMM convergence timer with iterations, solve time, 95th percentile and fallbacks; emergency compliance and junction-merge indicators; a spoofing view with broadcast and estimated positions; health and fleet-health panels with per-cell voltages and temperatures; and the DTC console. Runs can be exported and replayed without re-simulation. The demonstration is deployed at https://masooda.onrender.com/.

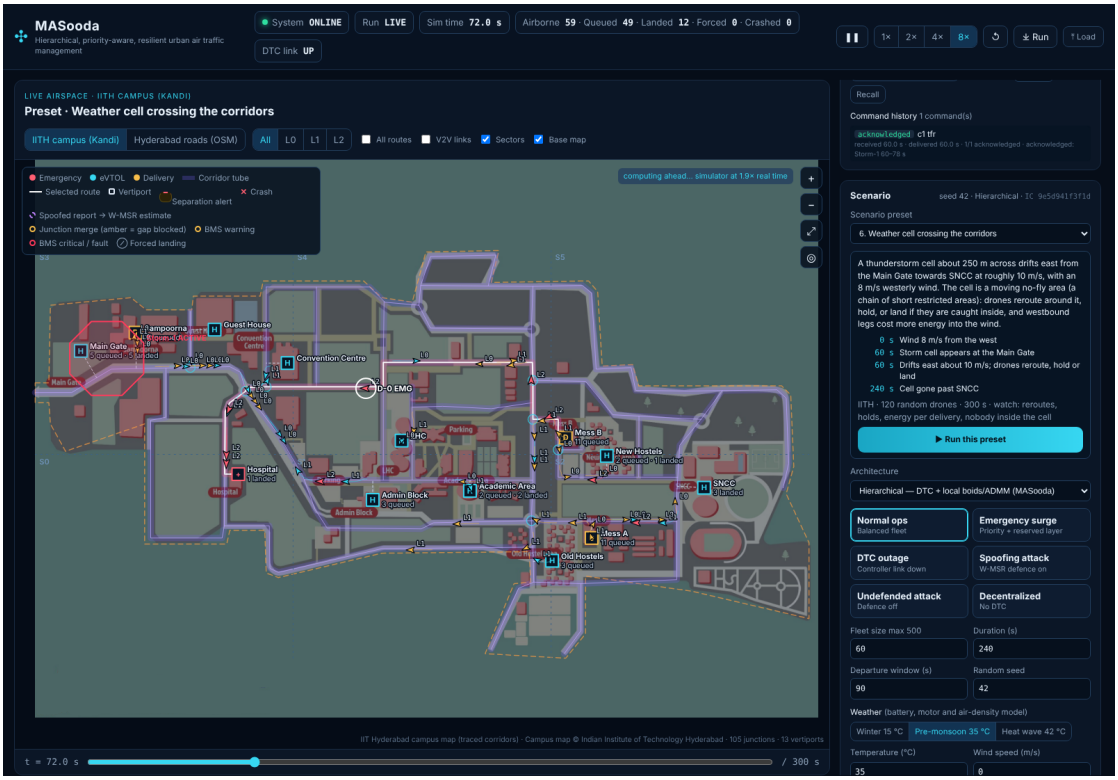


Figure 5.10: *MASooda Live Airspace on the IITH campus during the weather-cell scenario: drones colour-coded by type on road-aligned corridors with layer labels, a storm cell drawn as a restricted area over the Main Gate, vertiports with queue counts, and the scenario and architecture controls.*

For the corridor benchmark, a recorded dashboard (Figure 5.11) follows the deterministic demonstration scenario: corridor formation from 0 to 30 s, the first merges from about 30 s, an emergency agent throughout, a malicious agent from 60 s, the DTC link down

from 90 to 120 s and recovery to 180 s. It was rehearsed in three headless dry runs, two of them back to back, all without errors, and a fallback video was recorded from the same scenario.

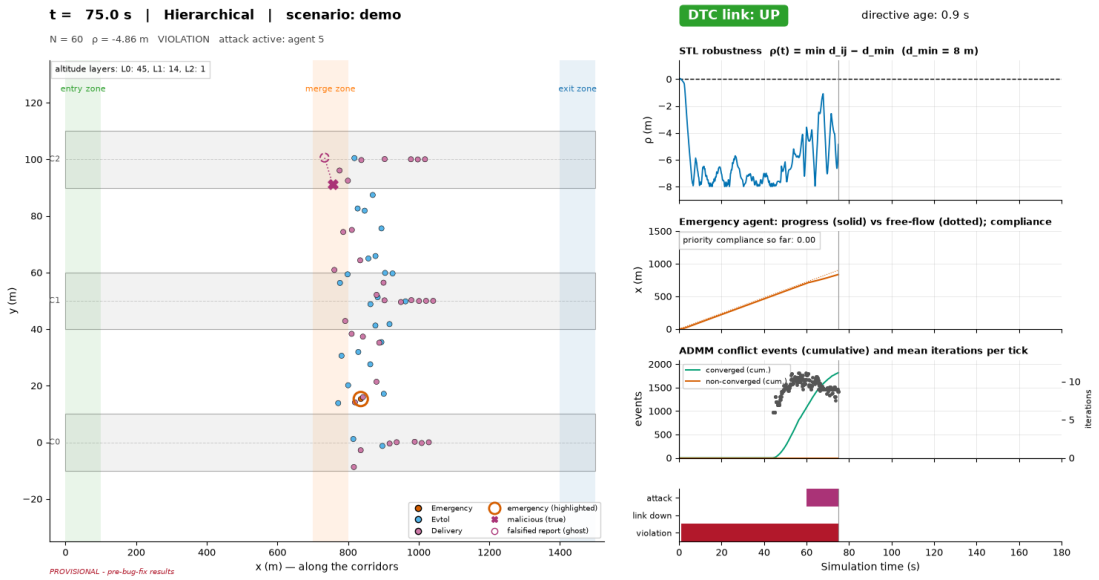


Figure 5.11: *Corridor-benchmark dashboard at $t = 75$ s: the malicious agent and its falsified "ghost" report, the STL robustness trace, emergency progress against free flow, cumulative ADMM events and iterations, and the attack, link and violation timeline.*

Summary against the Goals

Table 5.11 relates the results to the goals of Chapter 1.

5.2 Limitations

The following limitations should be kept in mind when reading the results. Most of them were identified by the team's own audit and are listed in the future work.

- **Provisional corridor results.** The Week 6 campaign used a horizontal separation metric that counts pairs on different layers, which dominates every hierarchical safety number. The metric and attack fixes were not merged before the campaign, and the campaign predates the seven-layer airspace and layer agreement; it has not been regenerated.
- **Statistical power.** Most groups use three seeds (five in E4), and 14 groups were flagged as weak. The attack effect in E4 is not statistically significant at this sample size.
- **ADMM is not a safety guarantee.** It is a one-step, linearised mitigation over detected pairs. Its prediction does not model the velocity-tracking lag and acceleration limit of the dynamics, and a non-converged group falls back to local rules.

Goal	Status	Evidence
2.5D simulation, 50–80 heterogeneous agents	met	30–75 agents in the benchmark; up to 500 in the live system
Boids local tier with merging	met	corridor and junction merging; merge metric to be revised
ADMM conflict resolution	met	$\approx$10 iterations independent of N, $<$ 2 ms/tick; mitigation, not a guarantee
Priority-aware DTC and degradation	met	emergency layer, priority order, cached directives; lost-link policies compared
Attack and W-MSR defence	met	range check without false flags; W-MSR estimate, trimmed price consensus, quarantine; effect not significant at 1 attacker in 60
Live STL monitoring	met	every tick, sound at samples; layer-aware in the live system
Architecture comparison	met, provisional	identical initial conditions; hierarchical safety numbers await the layer-aware re-run
Map-constrained airspace	met	IITH and Hyderabad networks, 29 of 30 verification runs without violations
Layer agreement	met	+36% arrivals, −71% ADMM events, 0 near misses at 300 drones
Drone health and console	met	10 scenarios end to end, no energy depletion

Table 5.11: *Status of the project goals.*

- **Idealised defence.** The trusted-range check uses an ideal range sensor computed from true positions, and in the corridor benchmark the W-MSR estimate is computed but local behaviour consumes the range-corrected reports. ADMM receives true intents, so it is not itself under attack. Only common-delivery falsification is implemented; per-receiver falsification is not.
- **Graph conditions.** The robustness condition under which W-MSR is guaranteed is not checked online, so the integrated resilience results are empirical.
- **Communication model.** Only the DTC link can fail. V2V messages have no latency, loss or staleness.
- **Capacity and metrics definitions.** The DTC counts overflow instead of throttling entry, and its capacity applies per (corridor, layer), whereas the E2 utilisation and E3 compliance metrics compare whole-corridor load with 15; the E2 merge-success and E3 compliance definitions therefore do not measure merging or prioritisation behaviour; and in the corridor benchmark arrival is counted by crossing the exit line rather than by reaching a destination.
- **Model fidelity.** The 2.5D kinematic model has instantaneous layer changes and no turn-rate, wind gusts or vehicle footprint. The battery and thermal models are simplified and not calibrated to a specific cell.
- **Map data.** Campus corridors were traced from a schematic map, and OpenStreetMap centrelines lack one-way tags. Corridors are a planning backbone, not approved airspace, and obstacles such as buildings and wires are not modelled.

6 Conclusion

6.1 Project Summary

MASooda set out to show that urban drone traffic can be managed by a hierarchy in which the drones resolve fast, local interactions among themselves while a Drone Traffic Controller takes slow, airspace-wide decisions, and that such a system can respect priorities, resist false information, survive the loss of its controller and verify its own safety online. Over six weeks the team built every component of that design (2.5D dynamics, V2V graphs, Boids local behaviour, priority-weighted consensus ADMM, a priority-aware DTC with a reserved emergency layer and cached directives, spoofing injection with a trusted-range and W-MSR defence, and an online STL monitor) and integrated them into one deterministic per-tick pipeline with shared interfaces and a single configuration. A reproducible 141-run campaign then compared the hierarchical design with centralised and decentralised baselines from identical initial conditions.

The campaign established that the local negotiation is cheap and predictable: ADMM needs about ten iterations regardless of fleet size, converges in every solve at the default limit, and costs under 2 ms per tick at 75 agents. It showed that an emergency agent saves 10–36 s compared with an identical ordinary agent while imposing only about ±2 s on others, that the cached-directive mechanism keeps the fleet flying through a 30 s controller outage, and that the undefended decentralised architecture, which uses raw reports, lost safety margin as the falsification offset grew (not statistically significant at five seeds). It also exposed, and this report documents, a horizontal separation metric that counts drones on different layers as conflicts, so the hierarchical safety numbers of that campaign remain provisional.

The project then moved from an abstract benchmark to a real place. On traced IIT Hyderabad campus roads and OpenStreetMap Hyderabad roads, with seven layers and a layer-aware metric, 29 of 30 verification runs with up to 90 drones had no loss of separation. A V2V layer-agreement protocol, in which drones negotiate layers by consensus dual ascent on congestion prices seeded by the DTC, raised arrivals by 36%, cut ADMM conflict events by 71% and removed all near misses in a 300-drone campus run. A battery,

BMS and thermal model for every drone, an operator console and ten scripted scenarios showed emergency right of way, lost-link policies, restricted areas, spoofer quarantine, heat-wave operation and health-driven diversions working end to end, with no energy depletion in the battery-failure and 60-minute operations scenarios. The system runs live in a browser with up to 500 drones.

The main lesson is that the hierarchy pays off most when each tier does what it is good at: the DTC is most useful as a price setter and arbiter of last resort, the drones as fast local negotiators, and safety is best judged by an independent monitor that uses the same notion of distance as the controllers.

6.2 Future Outlook

Making the evidence final. The first step is to rerun the corridor campaign with the layer-aware separation metric, the intact ADMM conflict components, a destination-based arrival test, revised merge and compliance definitions, and an outage scenario in which directives must change. Paired runs with more seeds (the shared campus HPC server makes 20 or more seeds per group practical) and pre-registered metrics would turn the provisional numbers into defensible ones.

Stronger safety. ADMM should predict the state that the dynamics will actually produce, including the velocity-tracking lag and acceleration limit, or be replaced by a short-horizon model predictive formulation or a control-barrier-function safety filter, so that separation can be guaranteed under stated assumptions rather than only mitigated.

Stronger resilience. The defence should use only information an aircraft really has (simulated noisy ranging rather than true positions), let the controllers consume the W-MSR estimate, extend the attack to per-receiver falsification and to ADMM messages, check the graph-robustness condition online, and model V2V latency, packet loss and staleness.

Health-aware traffic management. The DTC rule for unhealthy drones (Section 4.2.9) and energy-margin ADMM weights are already implemented; their evaluation belongs to the team's paper draft and is not reported here. The most original direction, and the focus of the team's planned IEEE paper, is to close the research gaps of Chapter 2:

- a *health-aware DTC* whose priorities and layer allocations use a health score built from energy reserve, BMS status and thermal margin, extending the existing BMS-status rule and pad reservation of Section 4.2.9 (G1, G2);
- *energy-fair ADMM*: the priority weights already scale with the energy margin in the map-constrained engine, so that drones with reserve absorb the manoeuvres; the remaining step is to measure the distribution of the energy cost of deconfliction (G3);
- *trusted health telemetry*, treating BMS reports like position reports, with plausibility checks against the power model and quarantine of drones that falsify health to gain priority (G4);

- an *STL energy and thermal shield*, with online robustness of $\mathbf{G}E \geq E_{\text{pad}}$ margin and $\mathbf{G}T_{\text{pack}} \leq T_{\max}$ triggering DTC actions before a violation (G5);
- a *hot-climate study* extending the existing comparison of health-aware and health-blind DTCs (nominal, low-charge, battery-fault and heat-wave conditions on the IITH map) with spoofed-health conditions, with energy-exhaustion events, diversions, emergency delay, throughput and fairness as metrics (G6).

Towards deployment. Longer term, the corridor network should be built from authoritative airspace data (Digital Sky zones, obstacle and building heights, temporary restrictions) with reservations in space and time, and the battery model should be calibrated against measured cell data. Corridors would then become candidate routes that can be checked against regulation, rather than a research backbone.

7 Contributions

7.1 Individual Contributions

The work was divided by module, following the project proposal, and every member's work was merged into one shared code base whose modules exchange data through a common data model. The contributions of each member are listed below.

P1 – Dhadheechi

Simulation world, safety specification and dashboard (shared with P6).

- The live simulation world of the corridor benchmark: agent state, 2.5D kinematics with speed and acceleration limits, and the deterministic per-tick update.
- The vehicle-to-vehicle communication graph (radius and k-nearest-neighbour variants) through which agents exchange position reports.
- Formalisation of the safety requirements as Signal Temporal Logic specifications, and the online STL monitor that computes the robustness of minimum separation at every step.
- Rendering of the benchmark runs and the metrics dashboard, including the robustness signal and the summary indicators used in the demonstration.
- Integration of the module outputs into the end-to-end benchmark runs.

P2 – Lalith Vishnu (Strategy and Conflict-Resolution Lead)

Consensus optimisation, ADMM and the map-constrained live airspace.

- **ADMM.** The consensus-ADMM conflict resolver: the solver, the safety and speed constraints, the convergence test, the adapter between the resolver and the simulation, conflict detection and N-agent conflict groups (the Weeks 1–4 work reached the shared repository through P5's integration commits). Its configuration parameters, best-iterate tracking, the E1 scaling experiment with a non-invasive profiler,

and the Week 6 fixes that keep connected conflict groups whole and fall back cleanly when ADMM does not converge.

- **2.5D airspace.** Proposed the multi-layer airspace and built the first seven-layer ADMM demonstration, together with the final demonstration run and its database-backed run storage.
- **Road-corridor simulator.** The map-constrained route engine, the road-network model, the spatial grid, corridor formation, and the priority and centralised modules for road maps; the IITH campus map traced by hand and the Hyderabad OpenStreetMap map.
- **Live demonstration.** The browser-based live sandbox (streaming server, compact frames, browser interface and run export), deployed publicly on Render; the DTC operator console; ten scenario presets; and the custom-drone builder.
- **Report defence and lost link.** The trusted-range check, which compares each neighbour's reported position with the physically sensed range and corrects a flagged report instead of discarding it; and the DTC link-outage handling, in which drones continue on their last cached directive, the staleness of that directive is tracked, and the operator can choose a lost-link policy.
- **Drone health.** The power model, battery model, BMS estimator, fault injection, landing-pad arbitration, and energy-margin ADMM weights; the V2V layer-agreement protocol.
- **Documentation and paper.** The full project audit, the integration plan, the research-gap review and the IEEE-style paper draft.

P3 – Bhuvan

Local behaviour: Boids rules and corridor merging.

- The Boids-style local rules of separation, alignment and cohesion that turn the neighbour picture into a desired velocity.
- Corridor following, with attraction towards the corridor centreline, and the corridor and platoon merging logic, including gap acceptance before a lane change and slowing down to open a gap.
- Tuning of the local-rule weights to keep pairwise separation above the minimum in the corridor benchmark, and the resulting autonomous formation behaviour.

P4 – Tanmay

Global tier: the Drone Traffic Controller.

- The centralised overseer node (DTC), which works from a coarse snapshot of the airspace instead of exact positions.
- Priority-aware corridor and layer allocation, in which agents are processed in priority order (emergency, eVTOL, delivery) under a per-corridor capacity.

- Priority arbitration across drone types using global traffic rules, and the issuing of directives to the agents over the DTC link.

P5 – Sampadram

Resilience: attack injection and resilient consensus.

- Fault and attack injection, in which malicious agents broadcast falsified positions with a constant or ramped offset from a set start time.
- W-MSR-style resilient consensus with trimming of extreme neighbour values, and the comparison against the undefended plain-averaging baseline.
- Integration of the members' Weeks 1–4 work into the shared repository.

P6 – Abdur Rehman

Simulation world, safety specification and dashboard (shared with P1).

- The live simulation world of the corridor benchmark: agent dynamics, the communication graph and the per-tick update.
- Formalisation of the safety requirements as STL specifications and implementation of the safety-specification monitor.
- Rendering and visualisation of the benchmark, and assembly of the final dashboard.
- Integration of all modules into the end-to-end pipeline.

References

Agnihotri, R. and S. K. Sharma (2025). *Optimized Conflict Management for Urban Air Mobility Using Swarm UAV Networks*. arXiv: 2512.12632.

Augugliaro, Federico, Angela P. Schoellig, and Raffaello D'Andrea (2012). "Generation of Collision-Free Trajectories for a Quadrocopter Fleet: A Sequential Convex Programming Approach". In: *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 1917–1922. DOI: 10.1109/IROS.2012.6385823.

Baculi, J. and C. Ippolito (2019). *Onboard Decision-Making for Nominal and Contingency sUAS Flight*. Tech. rep. NASA Ames Research Center. URL: https://ntrs.nasa.gov/citations/20190029195.

Bernardi, D., E. Pawlikowski, and J. Newman (1985). "A General Energy Balance for Battery Systems". In: *Journal of the Electrochemical Society* 132.1, pp. 5–12. DOI: 10.1149/1.2113792.

Bole, Brian, Matthew Daigle, and George Gorospe (2014). "Online Prediction of Battery Discharge and Estimation of Parasitic Loads for an Electric Aircraft". In: *European Conference of the Prognostics and Health Management Society*.

Boyd, Stephen et al. (2011). "Distributed Optimization and Statistical Learning via the Alternating Direction Method of Multipliers". In: *Foundations and Trends in Machine Learning* 3.1, pp. 1–122. DOI: 10.1561/2200000016.

Darrah, Taylor (2023). "A Health-Aware Replanning Framework for Unmanned Aerial Vehicles in Stochastic Environments". PhD thesis. Vanderbilt University.

Donzé, Alexandre and Oded Maler (2010). "Robust Satisfaction of Temporal Logic over Real-Valued Signals". In: *Formal Modeling and Analysis of Timed Systems (FORMATS)*. Vol. 6246. Lecture Notes in Computer Science. Springer, pp. 92–106.

European Union Aviation Safety Agency (2021). *U-space: Commission Implementing Regulation (EU) 2021/664 on a regulatory framework for the U-space*. URL: https://www.easa.europa.eu/en/domains/air-traffic-management/u-space.

Fainekos, Georgios E. and George J. Pappas (2009). "Robustness of Temporal Logic Specifications for Continuous-Time Signals". In: *Theoretical Computer Science* 410.42, pp. 4262–4291. DOI: 10.1016/j.tcs.2009.06.021.

Federal Aviation Administration (2020). *Unmanned Aircraft System (UAS) Traffic Management (UTM) Concept of Operations, Version 2.0*. U.S. Department of Transportation. URL: https://www.faa.gov/sites/faa.gov/files/2022-08/UTM_ConOps_v2.pdf.

Federal Aviation Administration (2023). *Urban Air Mobility (UAM) Concept of Operations, Version 2.0*. U.S. Department of Transportation. URL: https://www.faa.gov/sites/faa.gov/files/Urban-Air-Mobility-Concept-of-Operations-2.0.pdf.

Fiedler, Miroslav (1973). "Algebraic Connectivity of Graphs". In: *Czechoslovak Mathematical Journal* 23.2, pp. 298–305.

International Civil Aviation Organization (2005). *Annex 2 to the Convention on International Civil Aviation: Rules of the Air*. 10th ed. ICAO.

Jover, Jesús, Aurelio Bermúdez, and Rafael Casado (2022). "Priority-Aware Conflict Resolution for U-Space". In: *Electronics* 11.8, p. 1225. DOI: 10.3390/electronics11081225.

Kerns, Andrew J. et al. (2014). "Unmanned Aircraft Capture and Control via GPS Spoofing". In: *Journal of Field Robotics* 31.4, pp. 617–636. DOI: 10.1002/rob.21513.

Kleinbekman, I. C., M. A. Mitici, and P. Wei (2018). "eVTOL Arrival Sequencing and Scheduling for On-Demand Urban Air Mobility". In: *IEEE/AIAA 37th Digital Avionics Systems Conference (DASC)*.

Kopardekar, Parimal et al. (2016). "Unmanned Aircraft System Traffic Management (UTM) Concept of Operations". In: *16th AIAA Aviation Technology, Integration, and Operations Conference*. DOI: 10.2514/6.2016-3292.

KrisshnaKumar, P. et al. (2023). *Graph Learning Based Decision Support for Multi-Aircraft Take-Off and Landing at Urban Air Mobility Vertiports*. arXiv:2302.05849.

Kulkarni, Chetan, Edward Hogge, and Cuong C. Quach (2018). "Remaining Flying Time Prediction Implementing Battery Prognostics Framework for Electric UAVs". In: *AIAA Propulsion and Energy Forum*.

LeBlanc, Heath J. et al. (2013). "Resilient Asymptotic Consensus in Robust Networks". In: *IEEE Journal on Selected Areas in Communications* 31.4, pp. 766–781. DOI: 10.1109/JSAC.2013.130413.

Leishman, J. Gordon (2006). *Principles of Helicopter Aerodynamics*. 2nd ed. Cambridge University Press.

Maler, Oded and Dejan Nickovic (2004). "Monitoring Temporal Properties of Continuous Signals". In: *Formal Techniques, Modelling and Analysis of Timed and Fault-Tolerant Systems (FORMATS)*. Vol. 3253. Lecture Notes in Computer Science. Springer, pp. 152–166.

Ministry of Civil Aviation, Government of India (Oct. 2021a). *National Unmanned Aircraft System Traffic Management (UTM) Policy Framework*. Government of India. URL: https://www.civilaviation.gov.in/sites/default/files/migration/National-UTM-Policy-Framework-2021_24_Oct_2021.pdf.

Ministry of Civil Aviation, Government of India (2021b). *The Drone Rules, 2021*. URL: https://digitalsky.dgca.gov.in/assets/files/dronerules.pdf.

Montazeri, M., C. S. Kulkarni, and O. Fink (2024). *Prescribing Optimal Health-Aware Operation for Urban Air Mobility with Deep Reinforcement Learning*. arXiv:2404.08497. arXiv: 2404.08497.

Ni, Ruiqi, Zherong Pan, and Xifeng Gao (2022). "Robust Multi-Robot Trajectory Optimization Using Alternating Direction Method of Multiplier". In: *IEEE Robotics and Automation Letters*. DOI: 10.1109/LRA.2022.3159848.

Olfati-Saber, Reza (2006). "Flocking for Multi-Agent Dynamic Systems: Algorithms and Theory". In: *IEEE Transactions on Automatic Control* 51.3, pp. 401–420. DOI: 10.1109/TAC.2005.864190.

Olfati-Saber, Reza, J. Alex Fax, and Richard M. Murray (2007). "Consensus and Cooperation in Networked Multi-Agent Systems". In: *Proceedings of the IEEE* 95.1, pp. 215–233. DOI: 10.1109/JPROC.2006.887293.

Pant, Yash Vardhan et al. (2018). "Fly-by-Logic: Control of Multi-Drone Fleets with Temporal Logic Objectives". In: *ACM/IEEE International Conference on Cyber-Physical Systems (ICCPS)*, pp. 186–197.

Plett, G. L. (2004). "Extended Kalman Filtering for Battery Management Systems of LiPB-Based HEV Battery Packs: Part 1. Background". In: *Journal of Power Sources* 134.2, pp. 252–261.

Plett, Gregory L. (2015). *Battery Management Systems, Volume I: Battery Modeling*. Artech House.

Quinones-Grueiro, M. et al. (2020). "A Decision-Making Framework for Safe Operations of Unmanned Aerial Vehicles in Urban Environments". In: *Annual Conference of the Prognostics and Health Management Society*.

Rappaport, Theodore S. (2002). *Wireless Communications: Principles and Practice*. 2nd ed. Prentice Hall.

Reynolds, Craig W. (1987). "Flocks, Herds and Schools: A Distributed Behavioral Model". In: *Proceedings of the 14th Annual Conference on Computer Graphics and Interactive Techniques (SIGGRAPH)*, pp. 25–34. DOI: 10.1145/37401.37406.

Saha, Bhaskar et al. (2011). "Battery Health Management System for Electric UAVs". In: *IEEE Aerospace Conference*.

Salinas-Camus, Mariana, Chetan Kulkarni, and Marcos Orchard (2023). "Battery State-of-Health Aware Path Planning for a Mars Rover". In: *Annual Conference of the PHM Society*. Vol. 15. 1. DOI: 10.36001/phmconf.2023.v15i1.3511.

Scattolini, Riccardo (2009). "Architectures for Distributed and Hierarchical Model Predictive Control – A Review". In: *Journal of Process Control* 19.5, pp. 723–731. DOI: 10.1016/j.jprocont.2009.02.003.

Sharifi, Iman et al. (2026). *Separation Assurance between Heterogeneous Fleets of Small Unmanned Aerial Systems via Multi-Agent Reinforcement Learning*. arXiv: 2605.01041.

Sharma, Prashin and Ella Atkins (2021). *Prognostics-Informed Battery Reconfiguration in a Multi-Battery Small UAS Energy System*. arXiv: 2103.01883.

Stolaroff, Joshuah K. et al. (2018). "Energy Use and Life Cycle Greenhouse Gas Emissions of Drones for Commercial Package Delivery". In: *Nature Communications* 9, p. 409. DOI: 10.1038/s41467-017-02411-5.

Taye, A. et al. (2025). “Safe to Fly? Real-Time Flight Mission Feasibility Assessment for Drone Package Delivery Operations”. In: *36th International Conference on Principles of Diagnosis and Resilient Systems (DX)*. Vol. 136. OASIcs. DOI: 10.4230/OASIcs.DX.2025.8.

Treiber, Martin, Ansgar Hennecke, and Dirk Helbing (2000). “Congested Traffic States in Empirical Observations and Microscopic Simulations”. In: *Physical Review E* 62.2, pp. 1805–1824. DOI: 10.1103/PhysRevE.62.1805.

Yang, Xiao-Guang et al. (2021). “Challenges and Key Requirements of Batteries for Electric Vertical Takeoff and Landing Aircraft”. In: *Joule* 5.7, pp. 1644–1659. DOI: 10.1016/j.joule.2021.05.001.

Zhao, T. et al. (2025). “A Critical Review on the Battery System Reliability of Drone Systems”. In: *Drones* 9.8, p. 539. DOI: 10.3390/drones9080539.

Zongo, Alex and Peng Wei (2026). *eVTOL Aircraft Energy Overhead Estimation under Conflict Resolution in High-Density Airspaces*. arXiv: 2604.06093.

Appendices

A

Configuration Parameters

This appendix lists the parameters of the configuration shared by every module, grouped by component. Values are those used for the results in this report unless stated otherwise.

Table A.1: *Simulation, local behaviour, ADMM, DTC and resilience parameters.*

Parameter	**Value**	**Component**
Velocity-tracking time constant τ_v	0.5 s	dynamics
Detect-and-avoid sensing radius	30 m	map mode
Corridor centreline spacing	50 m	benchmark
Entry / merge / exit zones	0–100 / 700–800 / 1400–1500 m	benchmark
Separation radius r_{sep}	20 m	local behaviour
Weights w_g, w_{sep}, w_{ali}, w_{coh}	1.0, 35, 0.4, 0.1	local behaviour
Lateral gain k_{lat}	$0.5\,\text{s}^{-1}$	local behaviour
Look-ahead distance	40 m	local behaviour
Standstill gap d_0, time headway h	12 m, 1 s	car-following
Platoon radius	40 m	formation metrics
Merge slow-down k_v, merge gap, lateral speed	0.5, 15 m, 2 m/s	merging
Junction merge zone	40 m before a junction	map mode
Departure clearance at the junction	45 m	map mode
Crash radius (same layer)	2 m	map mode
ADMM safety margin factor	1.01	ADMM
ADMM relative tolerance	10^{-4}	ADMM
ADMM priority weights (P0, P1, P2)	10, 3, 1	ADMM
ADMM iteration sweep	5, 10, 20, 50, 100	E1
DTC sectors (map mode)	3×2 grid	DTC
Emergency layer	2 (benchmark), 6 (live)	DTC
Layer change gate	$3d_{\min}$ now, CPA $2d_{\min}$ in 10 s	DTC

Parameter	Value	Component
Tactical shift trigger, cooldown	miss $< d_{\min}$ within 3 s, 6 s	local tier
Command queue lifetime during outage	30 s	console
Attack modes	constant, ramp (1 m/s)	resilience
Liar counts in consensus tests	1, 2	resilience
Consensus steps (offline tests)	50	resilience
Layer agreement period	1 s	layer agreement
Weights w_f, w_p, w_e, w_s, w_{dir}	1.0, 0.6, 0.15, 0.25, 0.5	layer agreement
Battery climb factor k_b, reference climb	1.0, 30 m	layer agreement
Gain threshold, cooldown	0.15, 8 s	layer agreement
Price step α, target load κ, $\lambda_{\max}$	0.3, 2, 2.0	layer agreement
DTC price threshold θ	0.6	layer agreement
Stuck detection	$< 60\%$ cruise for 4 s	layer agreement
DTC settlement weights w_d, w_{pr}, w_{bat}, w_b, w_l, w_{sw}	1.0, 1.0, 0.5, 0.3, 0.5, 0.1	layer agreement
DTC look-ahead for settlement	80 m	layer agreement
Maximum fleet, maximum duration (web)	500 drones, 3600 s	live demo
Streamed frames per simulated second	5	live demo
Simultaneous live runs	2	live demo

Table A.2: *Drone health model: shared parameters and thresholds.*

Parameter	Value
Ambient temperature, site elevation	35 °C (heat wave 42–44 °C), 542 m
Climb speed	3 m/s
Usable energy floor	5% SoC
Random-fleet initial SoC, SoH	35–100%, 80–100% (seeded)
Resistance rise in the cold α_R	0.01 K^{-1} below 25 °C
Cell SoC and resistance spread	$\sigma = 0.4\%$, $\sigma = 3\%$
BMS period	0.5 s
SoC warning / critical	30% / 15%
Cell fault voltage	3.0 V
Pack warning / critical / fault	55 / 65 / 75 °C
Motor warning / critical	90 / 110 °C
Cell imbalance warning / critical	50 / 100 mV
Over-current warning	80% of the C-rate limit
Hysteresis	2% SoC, 3 °C, 10 mV
Response check period, reserve margin, landing time	1 s, 20%, 30 s
Overheat speed limit, critical hold before landing	60% of cruise, 30 s

Parameter	Value
Link: transmit power, PL_0, exponent, shadowing	27 dBm, 40 dB, 2.7, σ = 4 dB
Link: RSSI at 50% loss, slope	−95 dBm, 2.5 dB
GNSS: base HDOP, spoofing radius, gain	0.9, 150 m, 3.0
Faults: capacity loss, cell failure	30% charge lost; one cell −15% SoC and 4× resistance
Faults: overheating, motor, GNSS	60% of hover power as heat; motor health 0.8; HDOP 5
Battery service	swap 90 s or fast charge at 2C to 90%; 2 drones per pad
Repeated sorties	30 s turnaround; service below 35% SoC

	Delivery	Emergency	eVTOL
Empty mass / payload	6 kg / 2 kg	10 kg / 3 kg	450 kg / 100 kg
Rotors × diameter	4 × 0.38 m	6 × 0.55 m	16 × 1.5 m
Figure of merit / propulsive efficiency	0.65 / 0.80	0.65 / 0.80	0.70 / 0.85
Drag area / avionics power	0.08 m^2 / 25 W	0.12 m^2 / 40 W	1.2 m^2 / 500 W
Pack	6S, 22.2 V, 500 Wh	12S, 44.4 V, 800 Wh	108S, 400 V, 25 kWh
Pack resistance / C-rate limit	0.020 Ω / 5C	0.040 Ω / 5C	0.050 Ω / 5C
Pack heat capacity / hA_0 / hA_1	2.5 kJ/K / 3 / 1 W/K	4 kJ/K / 3 / 1 W/K	139 kJ/K / 60 / 25 W/K
Motor heat capacity / hA_0 / hA_1	0.4 kJ/K / 5 / 2 W/K	0.6 kJ/K / 6 / 2.5 W/K	20 kJ/K / 200 / 80 W/K
Cruise speed factor	0.9	1.2	1.0

Table A.3: *Drone health model: parameters per vehicle type.*

B

Scripted Scenarios

All scenarios use the IITH campus map, seed 42 and the default parameters unless stated. Scripted commands are applied at their tick exactly like operator commands, so every scenario replays identically. Table B.1 lists the vertiports.

Pad	Name	Role
V1	Hospital	hospital
V2, V3, V4	Main Gate, Guest House, Convention Centre	hub
V5, V6, V7	LHC, Admin Block, Academic Area	hub
V8, V10, V13	Mess B, Mess A, Sampoorna	depot (battery service)
V9, V11, V12	New Hostels, Old Hostels, SNCC	hub

Table B.1: *Vertiports of the IITH campus network.*

1. **Medical emergency during the lunch rush** (240 s, 120 drones, 70% delivery). A delivery wave leaves the messes and Sampoorna within 90 s. At 45 s an emergency drone carrying blood departs SNCC for the Hospital (2.3 km), and at 60 s a second one departs the Main Gate for the same pad (0.6 km). Focus: emergency compliance, delay against free flow and delay imposed on others.
2. **Peak-hour vertiport congestion** (300 s, 200 drones, 60 s departure window). Ground stops in the north-west and north-centre sectors from 20 s, released at 90 s and 120 s, and an 8 m/s speed limit in the south-centre sector from 30 s for 60 s. Focus: ground holds, pad queues, near misses and arrivals before and after release.
3. **Pop-up restricted area for a VIP visit** (240 s, 120 drones). A 360×270 m restricted area over the Academic Area and LHC roads is announced at 60 s, active from 70 to 180 s: drones inside are rerouted out or land, routes plan around it and departures from both pads are held.
4. **GPS spoofing near a restricted zone** (240 s, 100 drones). A restricted area around the Hospital approach is active throughout. Two spoofers broadcast positions offset 30 m towards the zone from 50 s; the defence corrects their reports, the map shows reported and estimated positions, and the operator quarantines both

at 110 s.

5. **DTC outage and lost-link procedures** (240 s, 100 drones, outage 80–150 s). Three variants with the fleet lost-link policy set to continue, hold or return to base. An operator command sent at 125 s is queued and delivered when the link returns at 150 s.
6. **Weather cell** (300 s, 120 drones, 8 m/s westerly wind). A storm cell about 250 m across drifts east from the Main Gate towards SNCC between 60 and 240 s, scripted as twelve overlapping restricted areas each active for 25 s.
7. **Non-cooperative intruder** (180 s, 80 drones). From 30 s a delivery drone without Remote ID flies from the Main Gate to SNCC. It is invisible to the DTC and to ADMM; other drones detect it only by onboard sensing within 30 m. A speed limit is set around its path at 60 s.
8. **Battery failure mid-route** (240 s, 100 drones). A delivery drone from Sampoorna to SNCC (2.4 km, 80% SoC) suffers a cell failure at 70 s; a second drone starting at 26% SoC has a capacity loss at 80 s. The operator grounds every CRITICAL drone at 85 s.
9. **Hyderabad heat wave versus a normal day** (300 s, 150 drones). Two variants at 44 °C and 30 °C with 2 m/s wind and two overheating faults at 60 s on long delivery routes.
10. **Sixty-minute operations with recharging** (3600 s, 60 drones). Drones fly repeated sorties with a 30 s turnaround; below 35% SoC a drone goes to a depot for a 90 s battery swap (two drones per pad, others queue). The fleet lost-link policy changes to return to base at 1200 s, and drones below 25% SoC are recalled at 2400 s.